

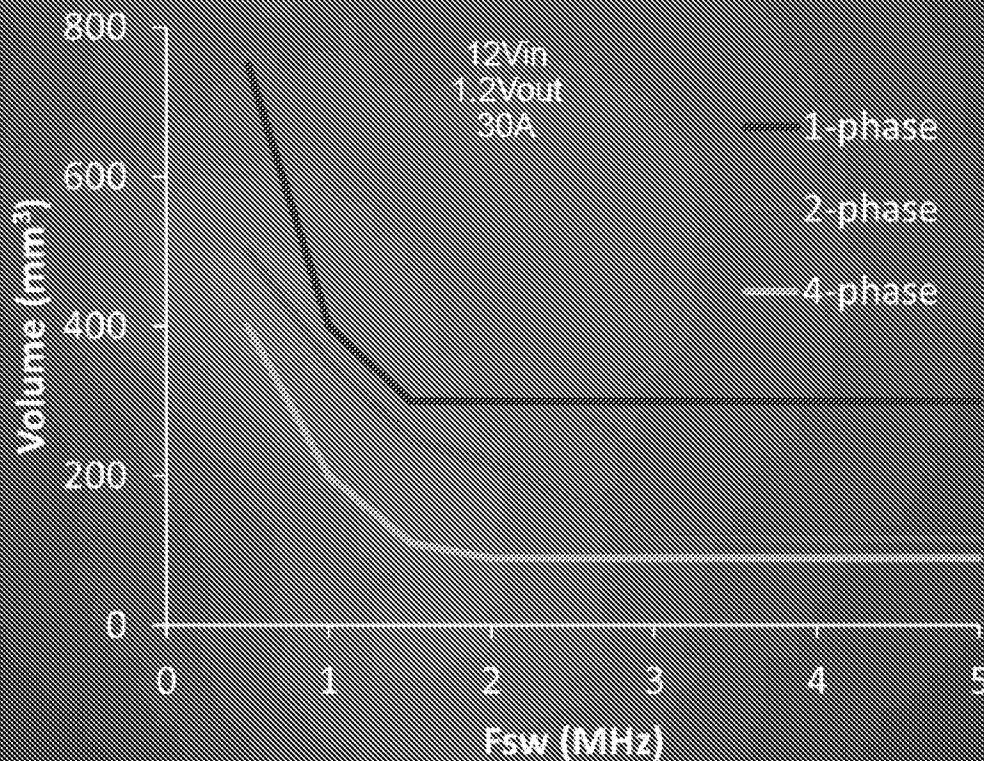
APPENDIX 2

Part 2 of 3

Challenges to Higher Density 12Vin POL solutions

(1) FET Switching Losses → Sarda GaAs FET

(2) High current inductors are large → Multiphase (lower I/phase)

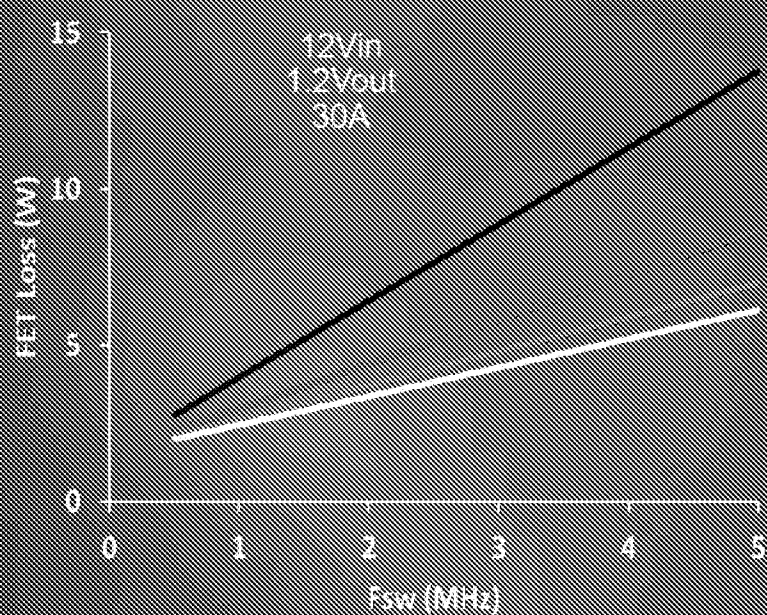


**Inductor Volume
decreases with higher
Phase count**

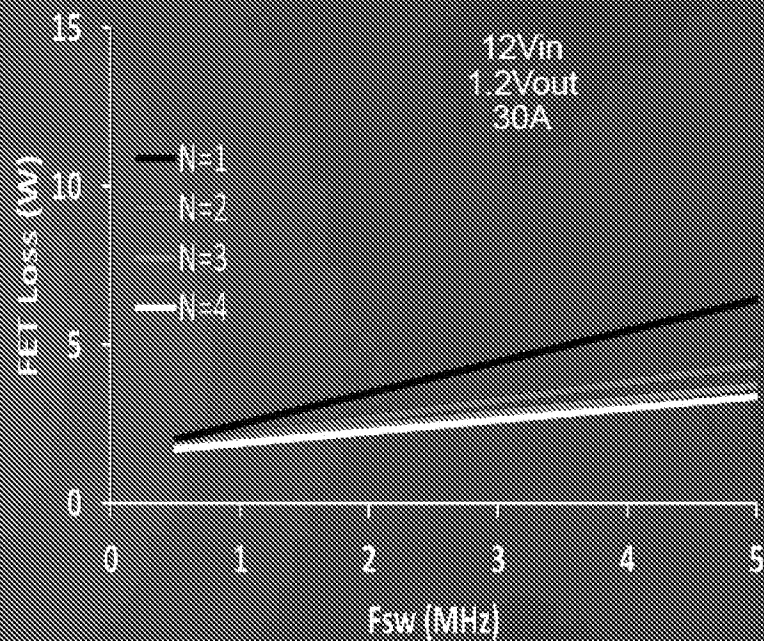
General example/trend
More than one inductor solution
choice per conditions examined

Another Multiphase Benefit

FET switching losses reduced



MOSFETs

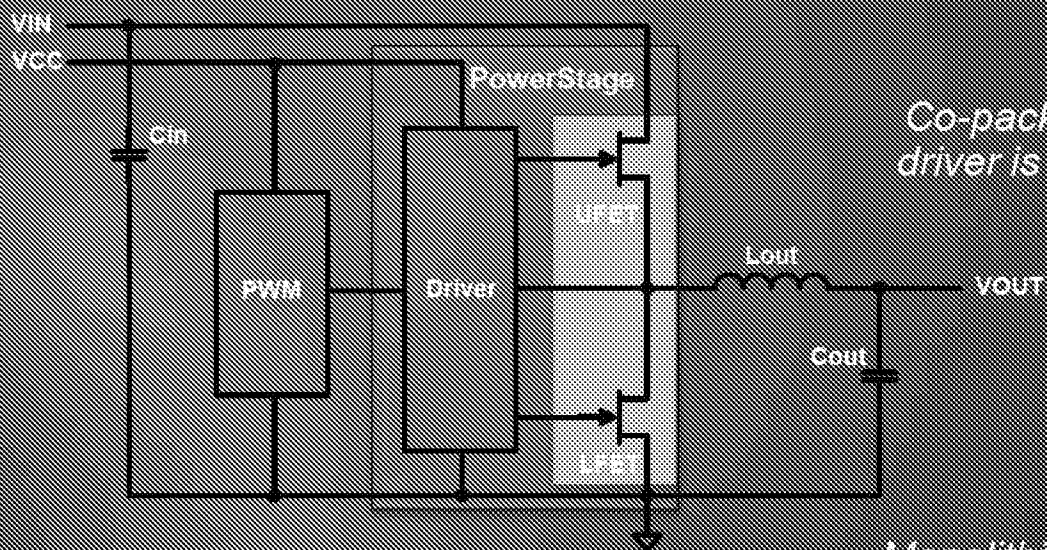


*Sarda GaAs FETs **

* In HIPS module (explained later)

Challenges to Higher Density 12Vin POL solutions

- (1) FET Switching Losses → Sarda GaAs FET
- (2) High current inductors are large → Multiphase (lower I/phase)
- (3) Parasitic Impedances

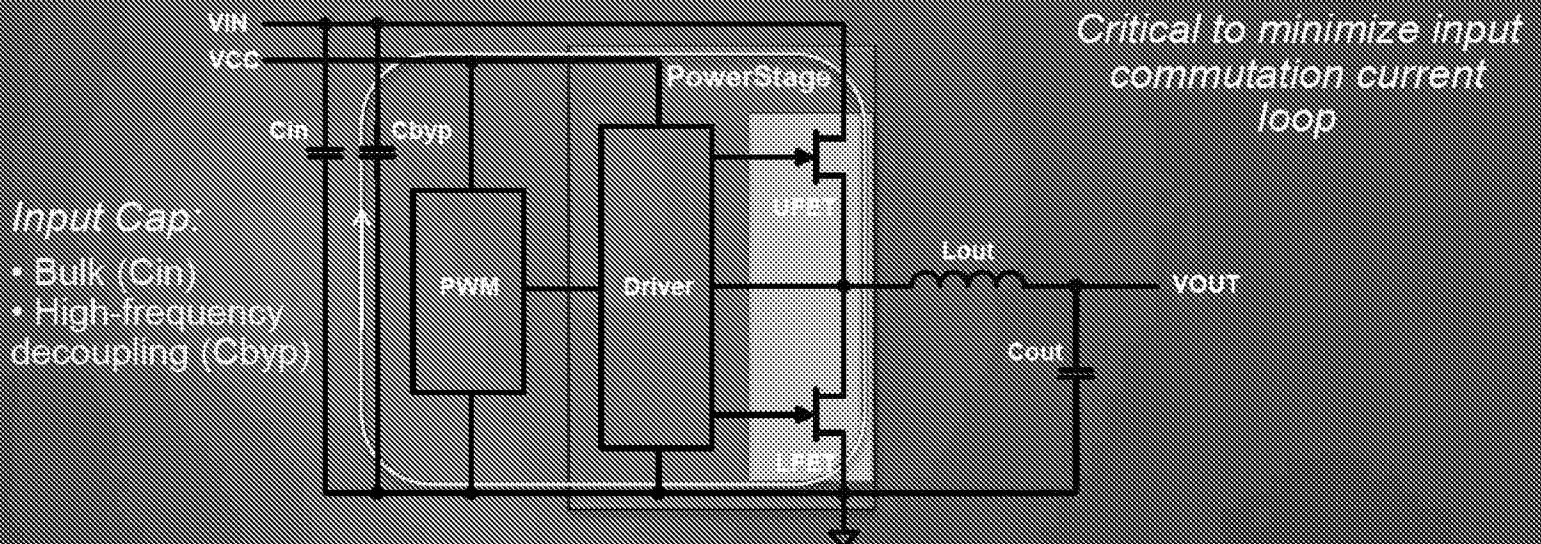


Co-packing FETs and driver is a requirement

Monolithic UFET / LFET minimizes critical common source inductance

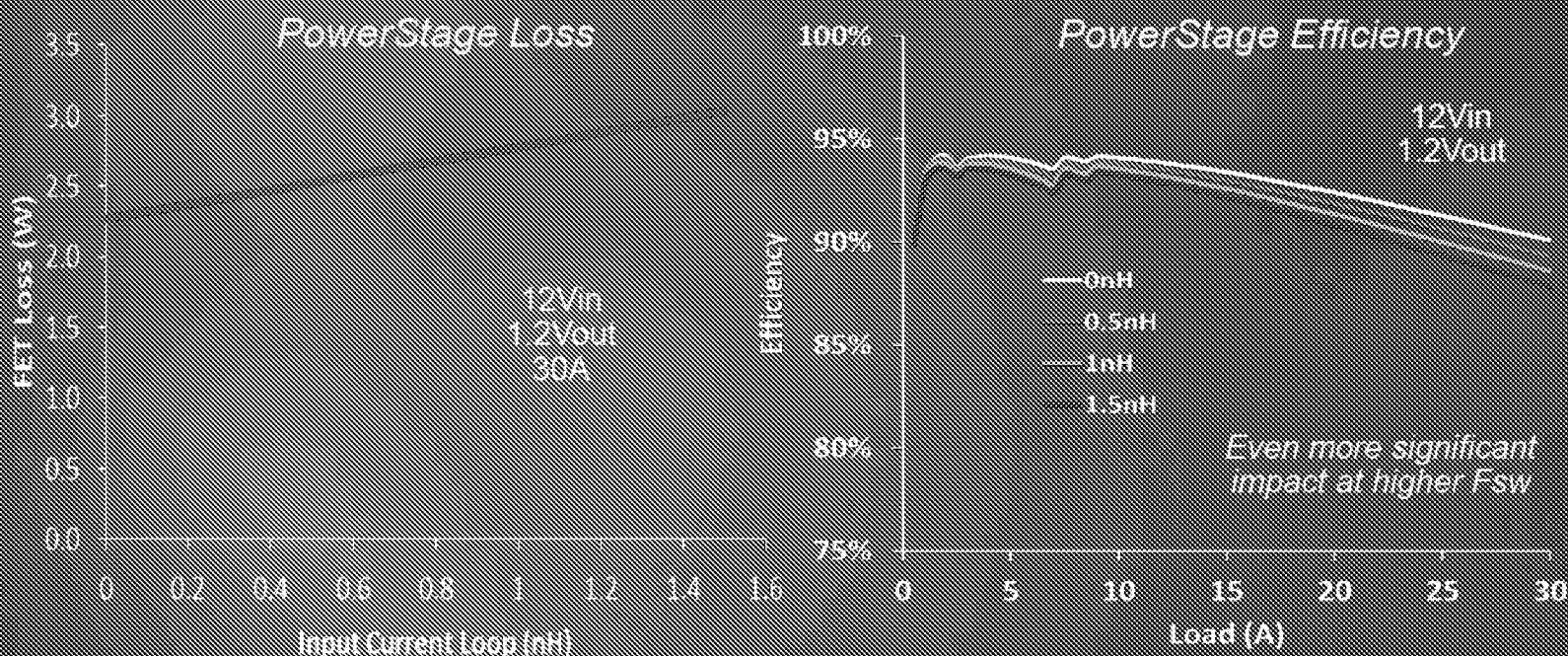
Challenges to Higher Density 12Vin POL solutions

- (1) FET Switching Losses → Sarda GaAs FET
- (2) High current inductors are large → Multiphase (lower I/phase)
- (3) Parasitic Impedances



Minimize Input Current Loop

Quantifying with Sarda Gen1 HIPS (4-phase, 2MHz)

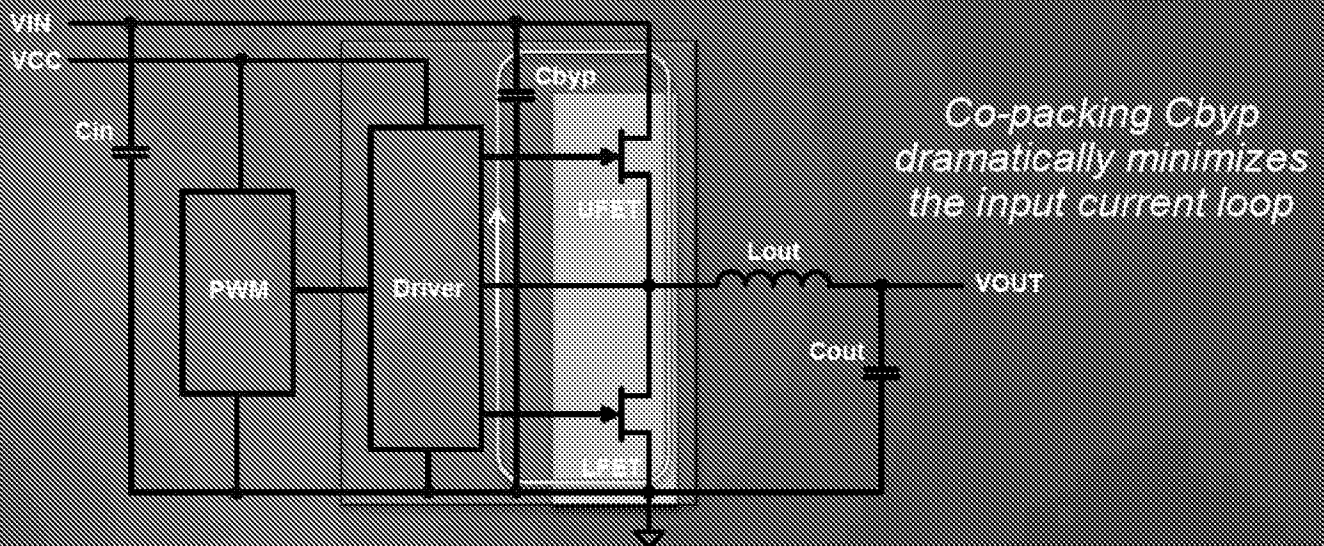


Additional Information on impact of Input Current Loop Inductance:

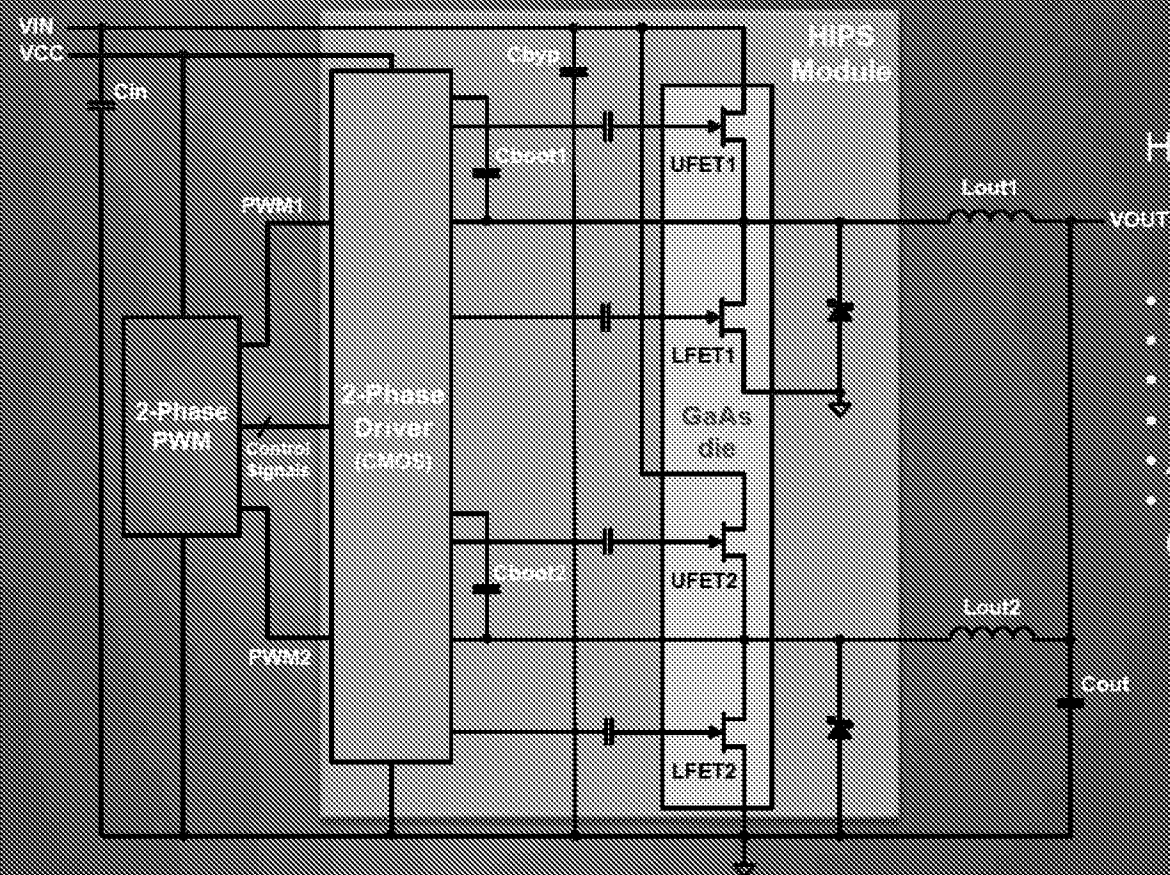
- APEC 2013: "Understanding the Effect of PCB Layout on Circuit Performance in a High Frequency Gallium Nitride Based Point of Load Converter", David Reusch, et al.
- IMAPS 2013: "A Highly-Integrated GaAs-based Module for DC-DC Regulators", Greg Miller, et al.

Challenges to Higher Density 12Vin POL solutions

- (1) FET Switching Losses → Sarda GaAs FET
- (2) High current inductors are large → Multiphase (lower I/phase)
- (3) Parasitic Impedances → Integrated Power Stage



Integrated Power Stage Approach: HIPS



HIPS: Heterogeneously Integrated Power Stage

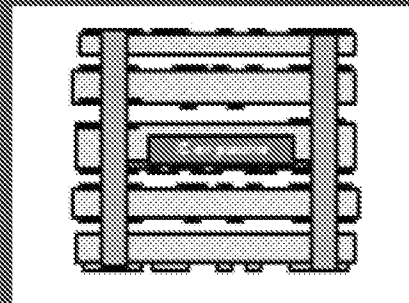
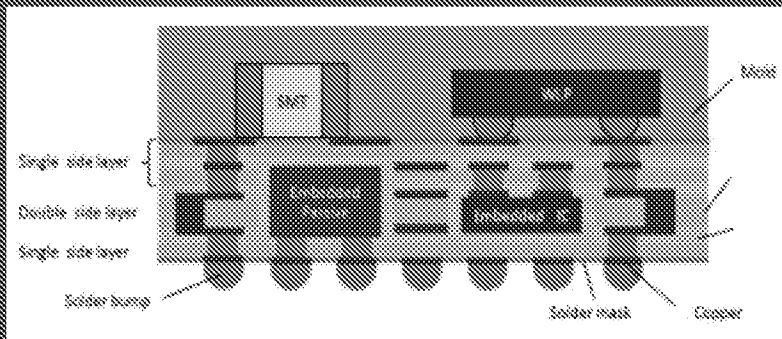
- Monolithic Quad GaAs die
- 2-phase CMOS driver
- C_{by}
- Boot caps
- AC-coupled gate caps
- Schottky diodes (for diode emulation)

Integrates performance-critical components

HIPS Module

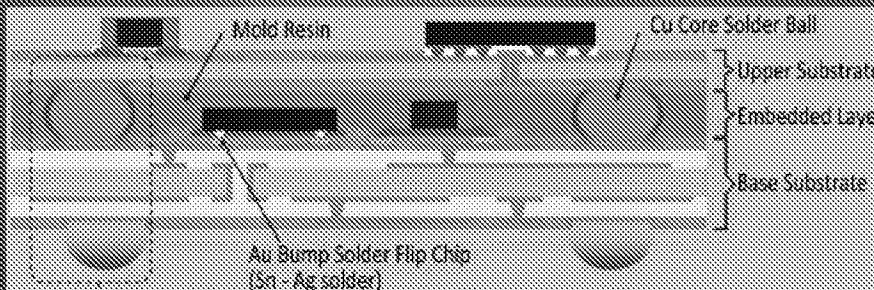
Available technology to enable compact multi-chip HIPS module:
Embedded Die in Substrate

Two motivations: (1) Performance (*minimize parasitics for multi-MHz operation*)
 (2) PowerStage package size

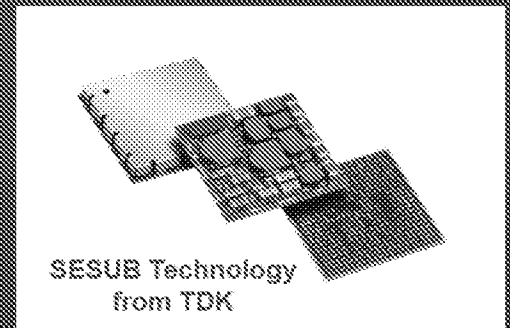


ECP®
 Technology
 from AT&S

Chipset™ Technology from FCI and Fujikura



MCeP® Technology from Shinko Electric

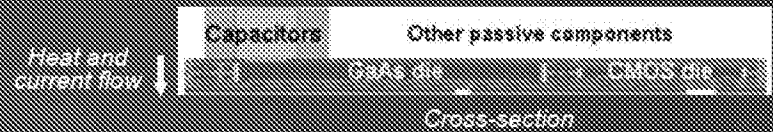


SESUB Technology
 from TDK

HIPS Approach

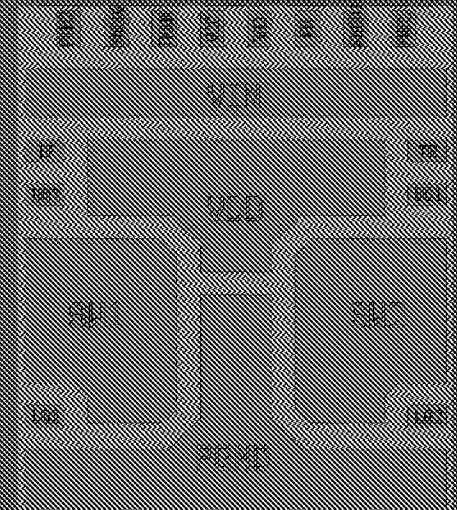
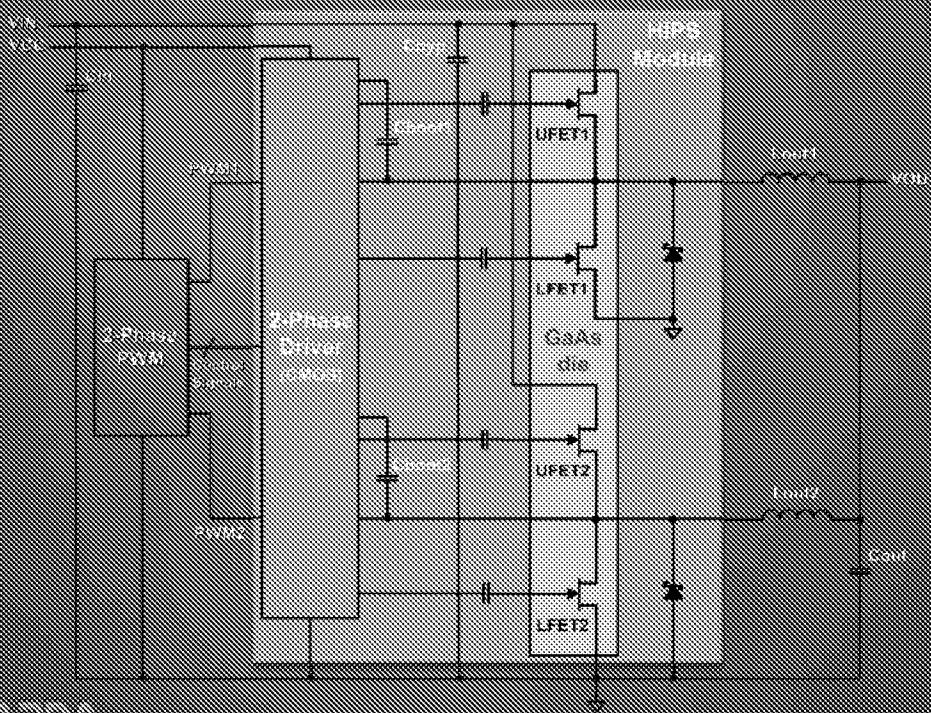
Embed Active components

- Quad GaAs FET die
- CMOS 2-phase driver



Integrate performance-critical components

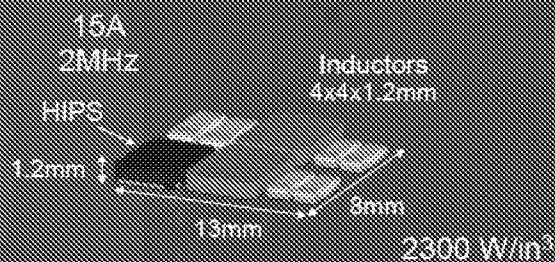
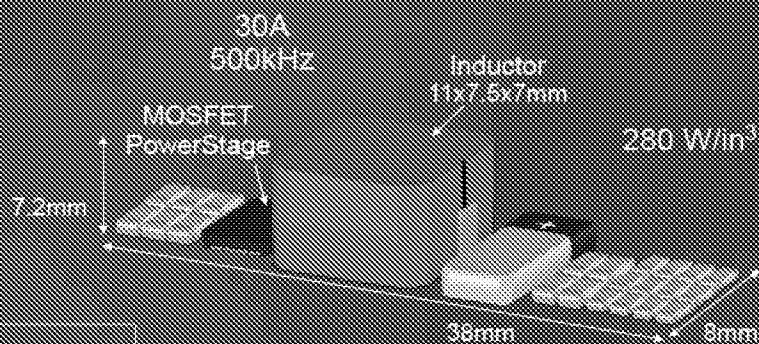
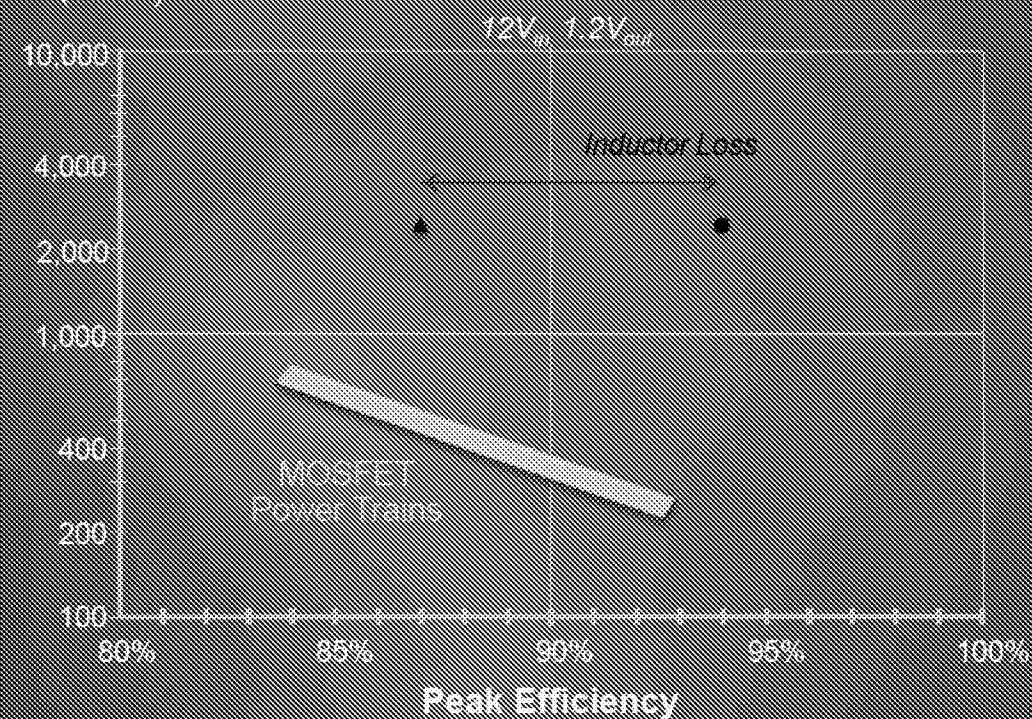
- Driver & FET
- Cbyp, Boot caps, gate caps, Schottky's



5x4x1mm QFN package
15A capability
1-5MHz

HIPS Performance

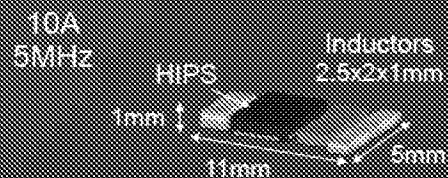
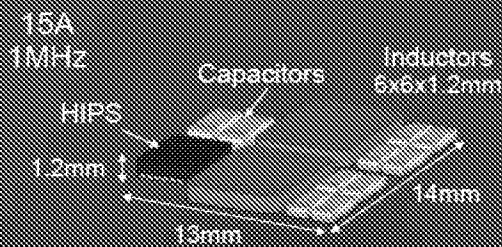
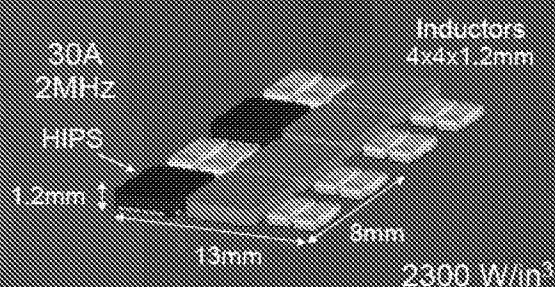
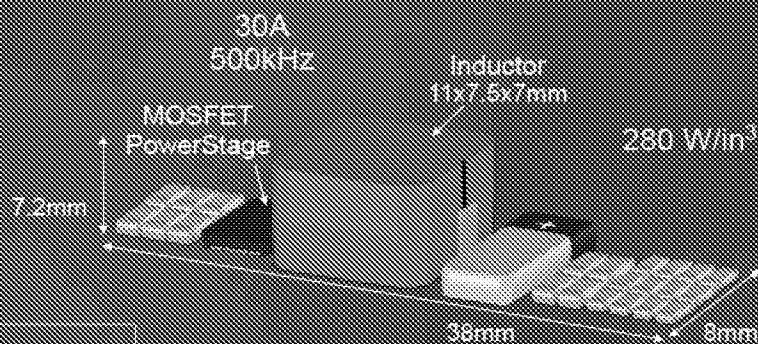
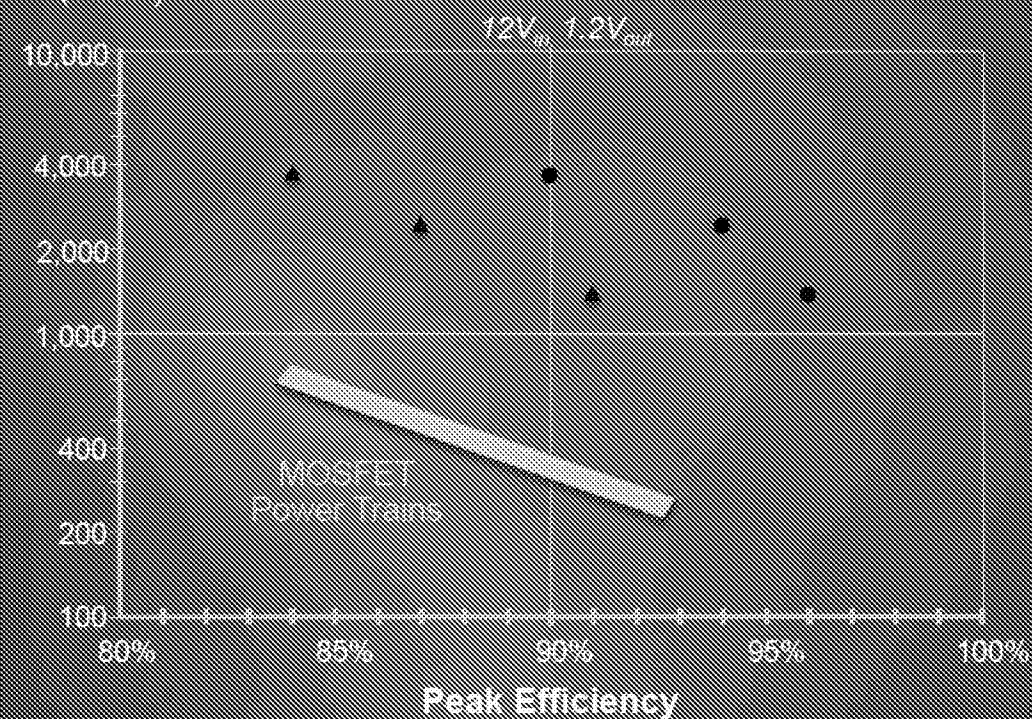
Power Train
Density
(W/in³)



- Power Stage efficiency
- Power Train efficiency

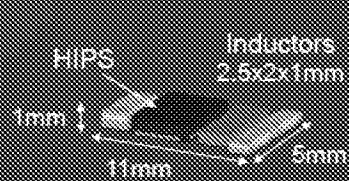
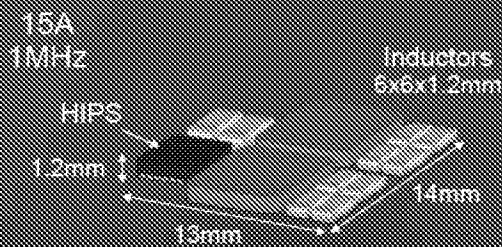
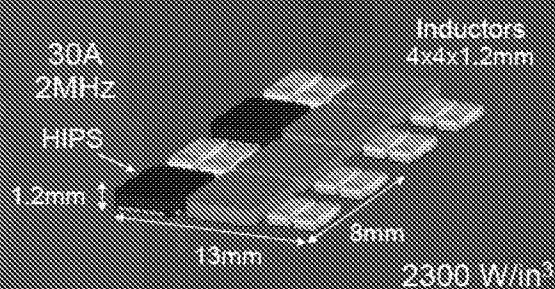
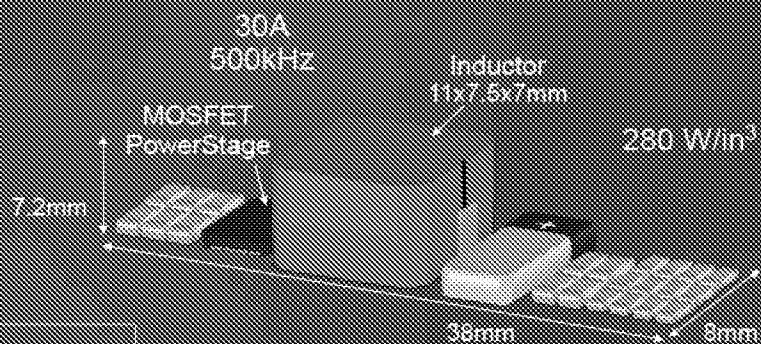
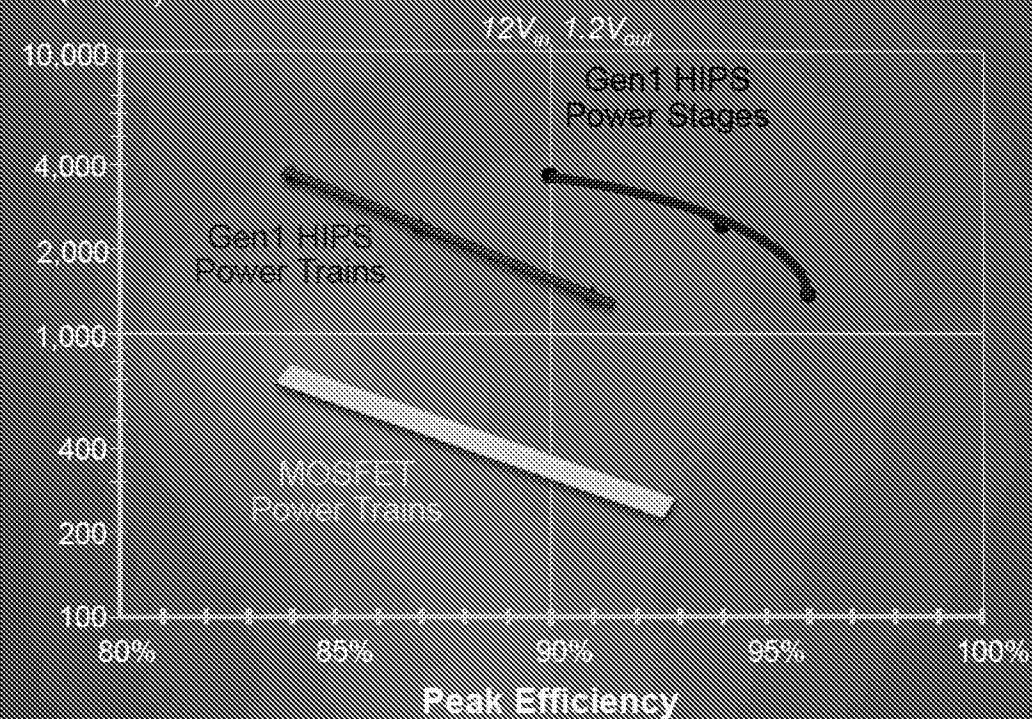
HIPS Performance

Power Train
Density
(W/in³)

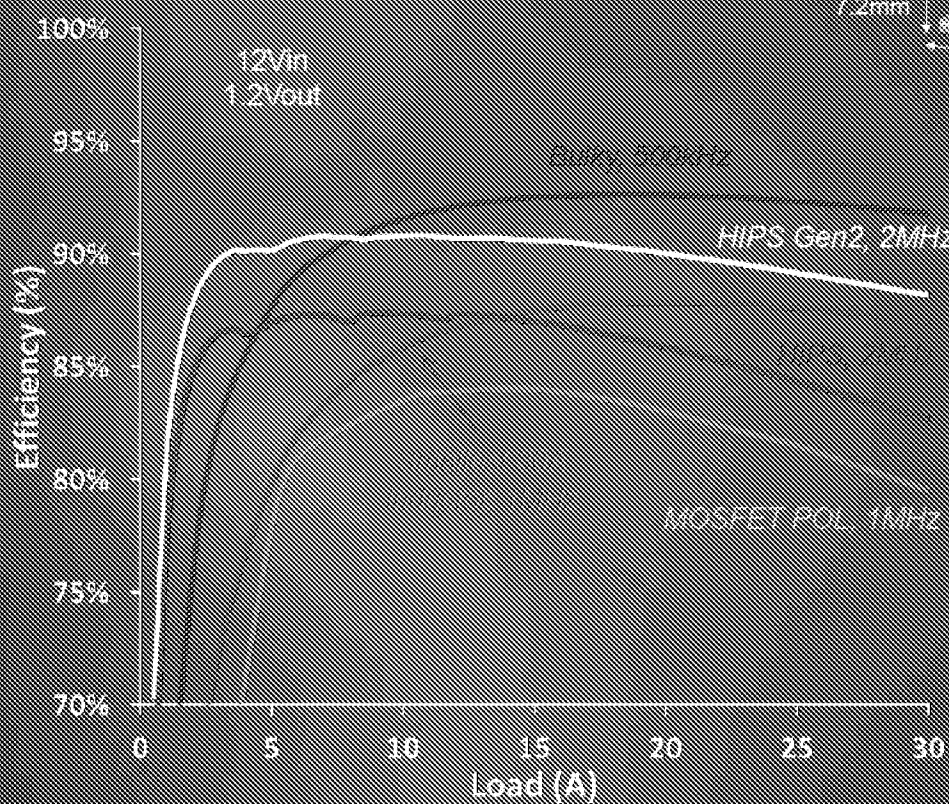


HIPS Performance

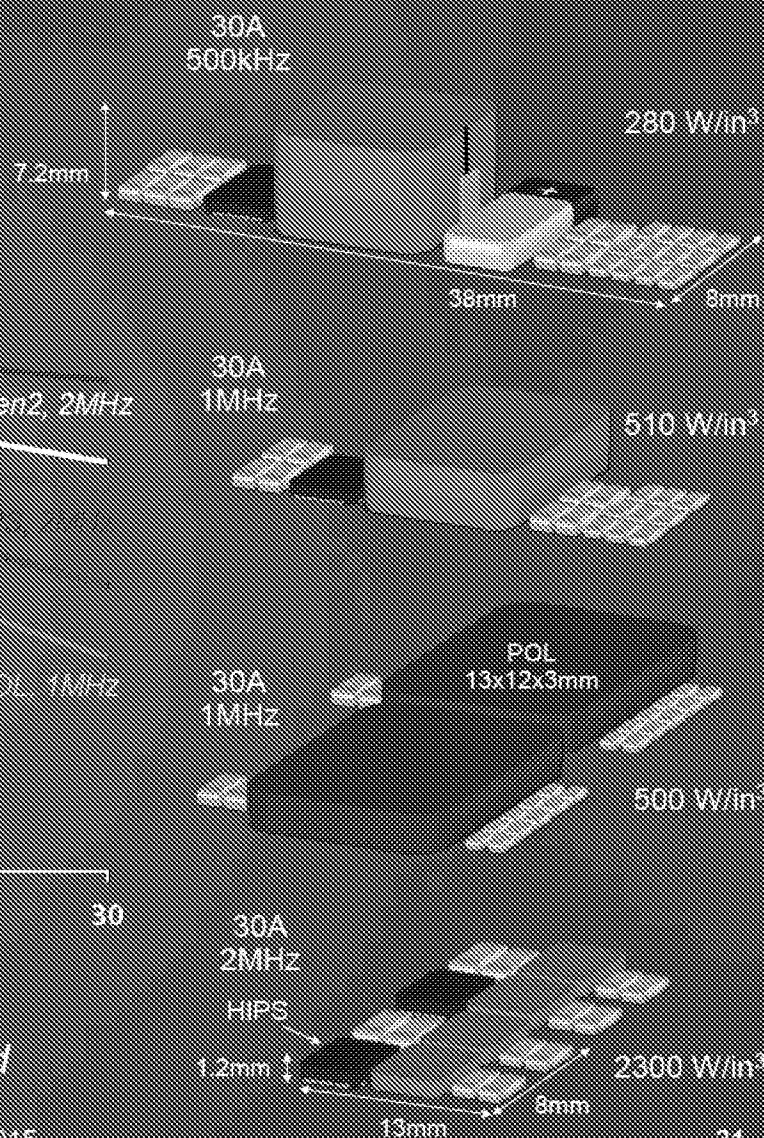
Power Train
Density
(W/in³)



Efficiency Examination

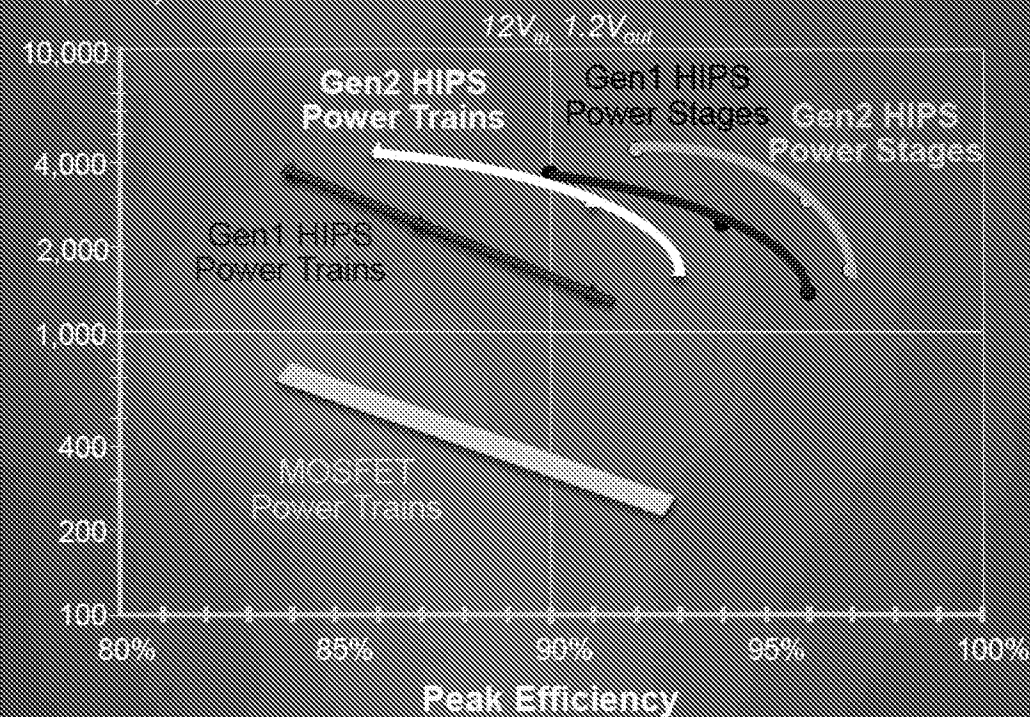


Gen2: Ron-A FOM halved
Rdson-Qg FOM halved

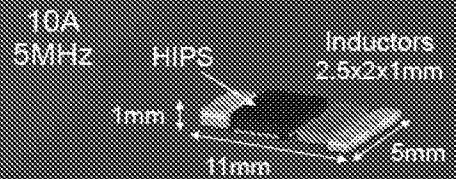
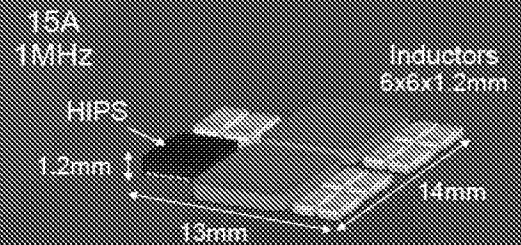
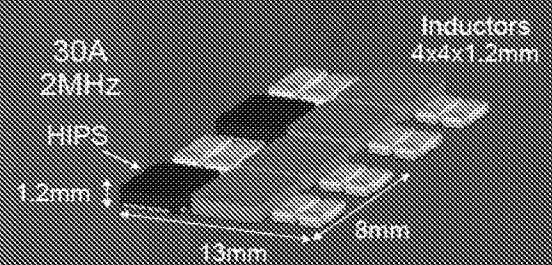
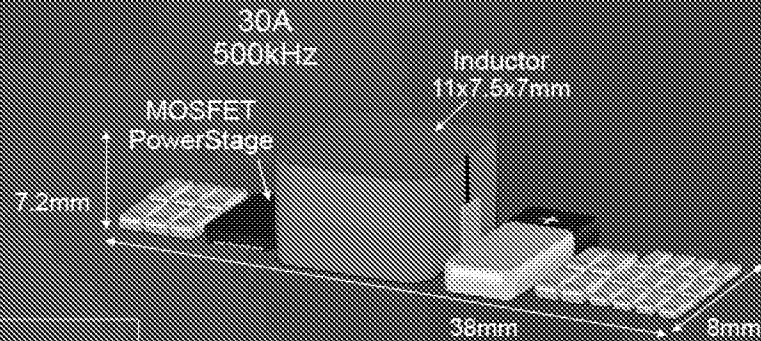


Gen2 HIPS Performance

Power Train
Density
(W/in³)



Examines Gen2 performance in 1-5MHz range
Can push to higher F_{sw} also



Summary

- **HIPS Module**

- Integrate performance-critical components
- Dramatically increase F_{sw} to shrink the output filter

- **Embedded Die in Substrate (3D Packaging) can enable new levels of power density in 12Vin POLs**

- **Compound semiconductors required to fully realize benefits**

- GaAs is ideal for 12V input

- **Enable higher switching frequencies**

- Up to 5MHz is only a starting point → HIPS roadmap to 10's of MHz

A 400 Amp fully Integrated Silicon Voltage Regulator with in-die magnetically coupled embedded inductors

J. Ted DiBene II Ph.D.,

P.R. Morrow Ph.D., C. - M. Park Ph.D., Henry
W. Koertzen Ph.D., Peng Zou Ph.D., Fenardi
Thenus, Xiaobei Li Ph.D., Stephen W.
Montgomery Ph.D. Ed Stanford, Robert Fite,
Paul Fischer Ph.D.

- Intel Corporation

For APEC 2010 February 25th in Palm Springs CA



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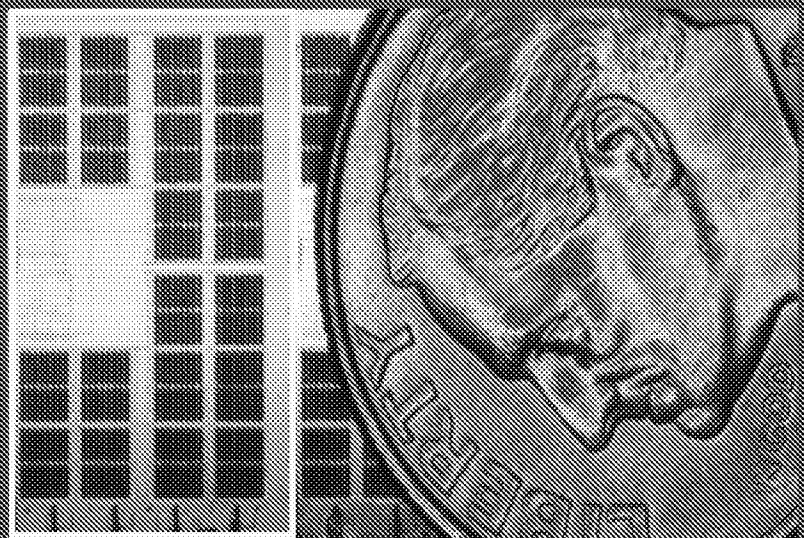
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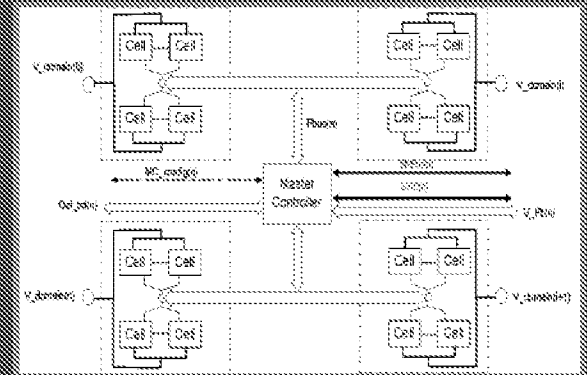
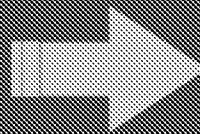
Agenda

- Architecture
- Circuits
- Results
- Conclusion

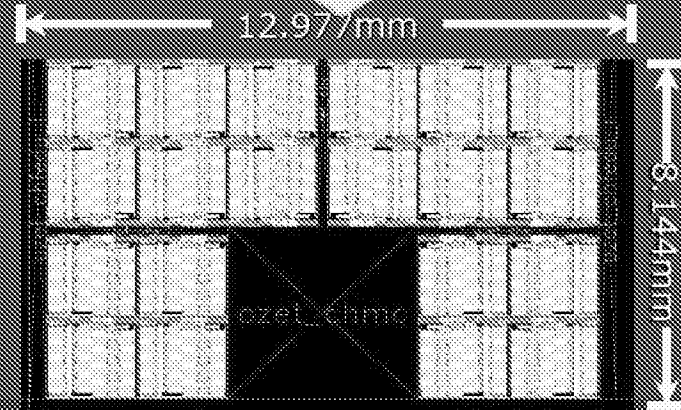


Integrated VR Technology

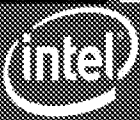
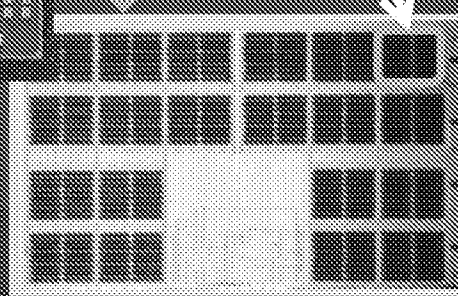
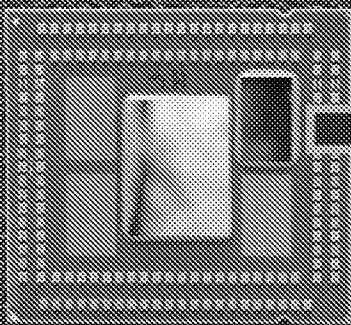
- 'Common Cell' Architecture – 20 cells
- Architecture supports flat efficiency curve
- Fine grain power management
 - Allows for multiple voltage rails
- Telemetry and Margining features
- Active Voltage Positioning for current sharing and balance
- Control features, including: JTAG, FPGA, Test/BIST



General Arch

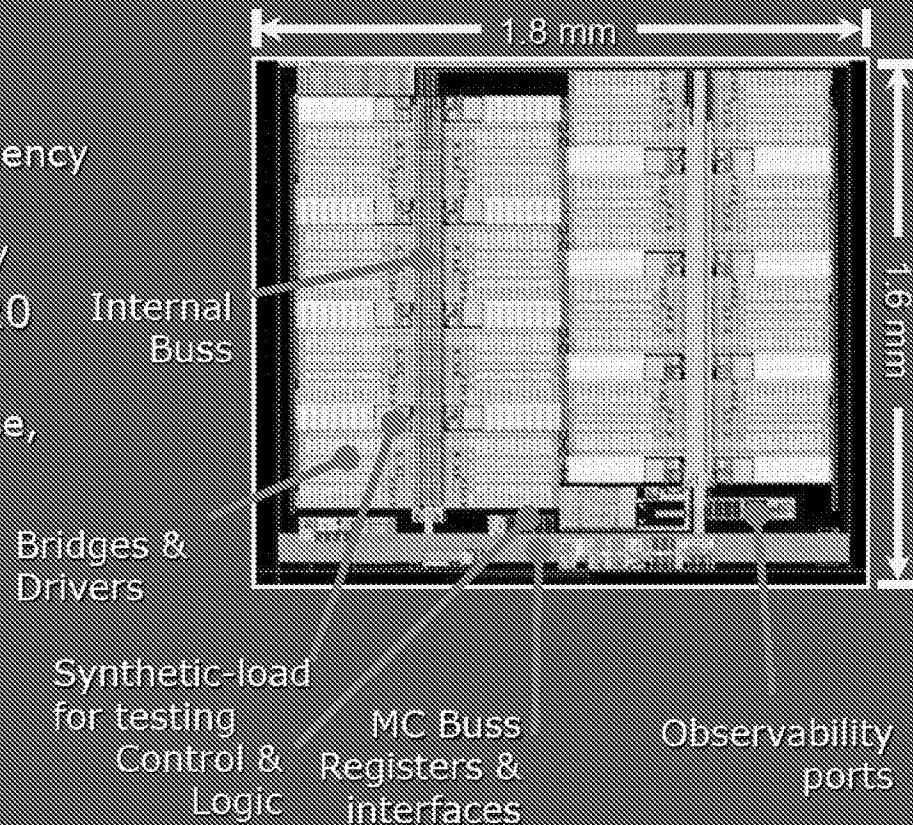


*Power cell
- 2.8 mm²*



Review: Power Cell Architecture

- Each Power cell = Mini VR
 - Up to 25A rating* - tested
 - Programmable switching frequency 30MHz to 140MHz
 - Ring coupled inductor topology
- 16 phases per power cell, 320 phases per chip
 - High phase count reduces noise, ripple
 - High granularity
 - Cell shedding
 - Bridge shedding
- BIST
 - Self-load and characterization system.



* Thermally constrained

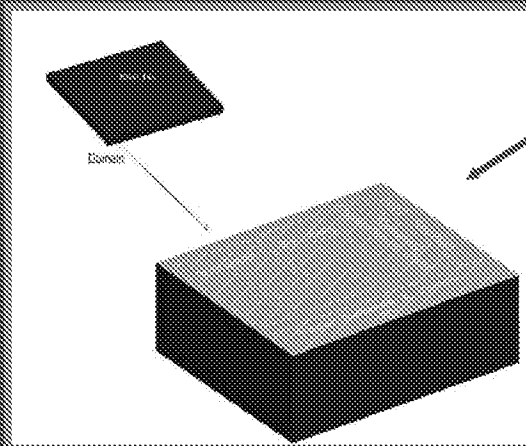


Agenda

- Architecture
- Circuits/Magnetics
- Results
- Conclusion



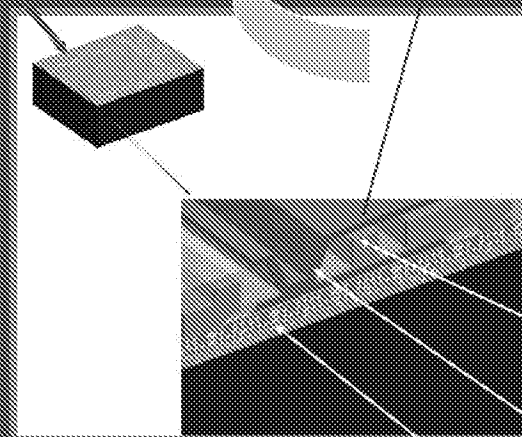
Thin-Film On-die Magnetics



Domain



Magnetic Cross section



L metal

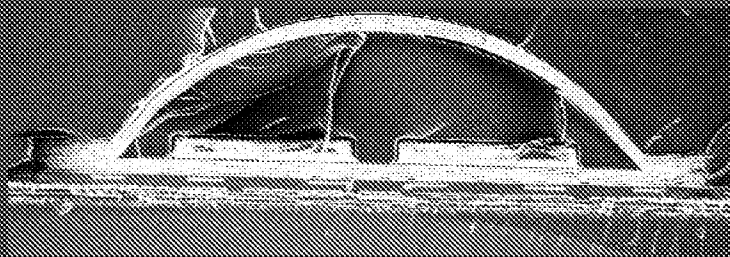
$\text{Ni}_{80}\text{Fe}_{20}$

m1-mX

- Technology targets & Stackup
- 90 nm technology for test devices
- 7-8 metal layers + thick metal(C4) + 2 Magnetic layers
- M1-m4/5 routing – bridge connections – decoupling capacitors
- M8 – inductor/transformer interconnect – $\sim 10 \mu\text{m}$ thick metal
- M7,M9 –magnetic material layers– $\sim 4 \mu\text{m}$ thick (laminated) $\text{Ni}_{80}\text{Fe}_{20}$



Gains from on-die magnetics



Magnetic SEM Cross section

- Energy density increased
- Volume shrinks
- Power Loss decreased



$$W_M = \frac{1}{2} \iiint_V \vec{B} \cdot \vec{H} dv \cong \frac{B^2 v}{2\mu_r \mu_0}$$

$$\frac{W_M}{W_A} \cong \mu_r$$

$$W_a \cong W_m \Rightarrow v_m \mu_r \cong v_a$$

$$\frac{R_a}{R_m} = \frac{P_a}{P_m} \approx \frac{l_m \sqrt{\mu_r}}{l_a}$$

Energy density in thin film magnetics volume compared with air core inductor is proportional to permeability μ_r which is typically > 1000

Thin-Film Magnetics in relation to VR Ckts

- $L_{\text{self}} \sim 17\text{nH}$ per phase. - $K \sim 93\%$
- 16 phase - 25A/cell I_{max}
- 2.8 mm^2 per cell

$$P_{L_c} \cong \Delta I^2 R_c = \frac{\Delta I^2 l_c}{\sigma w_c t_c}$$

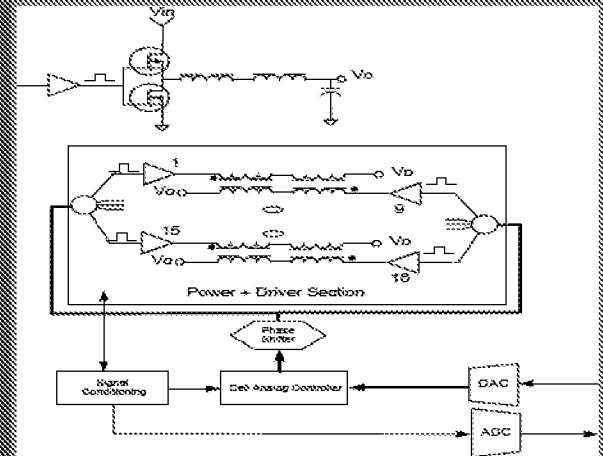
$$L \cong \frac{\Delta V \Delta T}{\Delta I} = \frac{\Delta V (1/F) D}{\Delta I}$$

$$* L_{\text{die}} \cong \frac{\Delta V (1/F) D}{\Delta I} \cong \frac{B_{\text{Sat}} w t_m}{\Delta I}$$

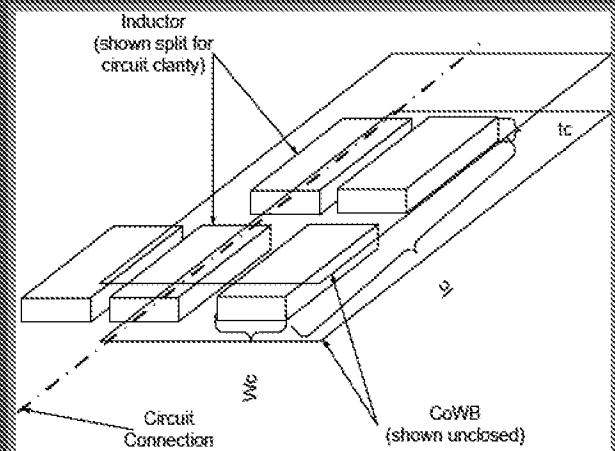
Single Magnetic Top View



Cell Circuit



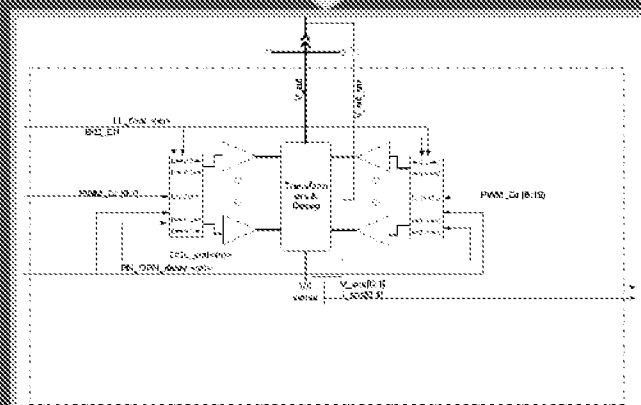
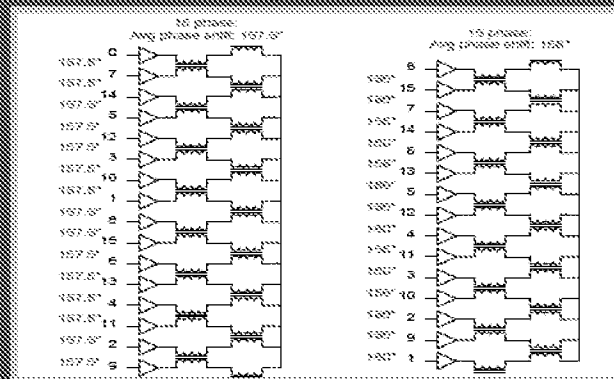
Inductor Physical



Power Train Architecture

- Cell level power train & Local Controller
- 16 phase 60-140 Mhz (per phase) coupled inductor
- Controller – type I analog
- Current Sense
- Flat efficiency with bridge shedding
- Loop programmable
- Register control between master controller and local cell controller
- Monitor and Observability thru pass-gate port design

16 phase VR



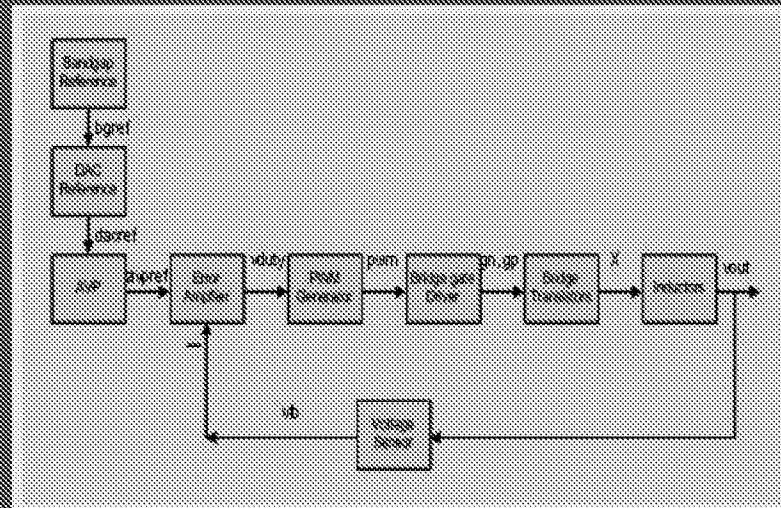
Power Cell Power Train Block



Power Cell Circuits

- VR Loop
- Feedback Control
- PWM & Phase generation
- Bridge driver and Transformer
- Reference
- Bandgap with wideband PSR
- VID control D/A converter
- Linear regulators for noise isolation
- Other features
- Synthetic loads – support test activities
- I, V, T sensors

VR Loop circuits

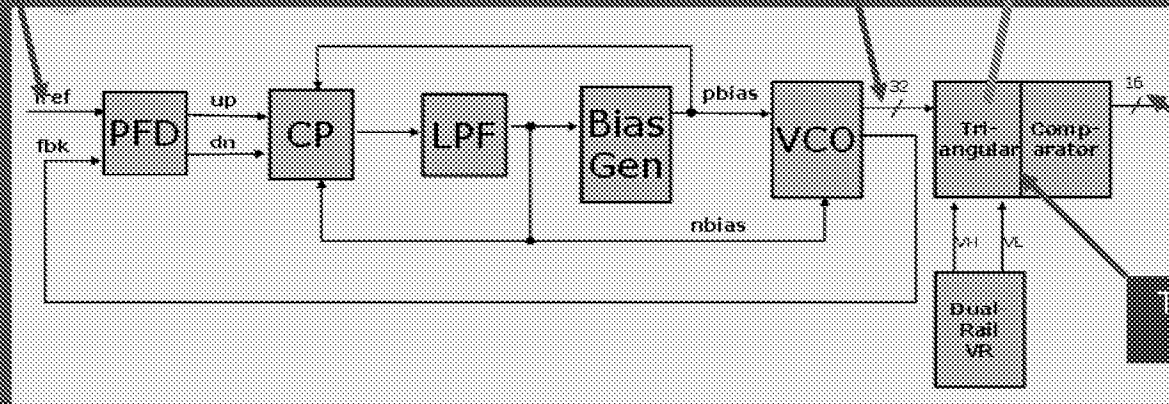
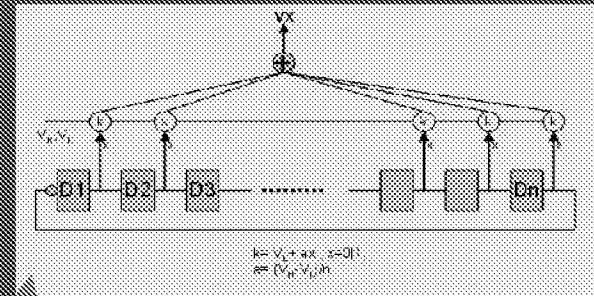


PWM topology

- Differential low power self-biased PLL
- Pulse width control by:
 - Clock => Triangle wave => pulses with variable duty cycle

Ref Clock
1 phase
50% duty

Clock
16 phases
50% duty

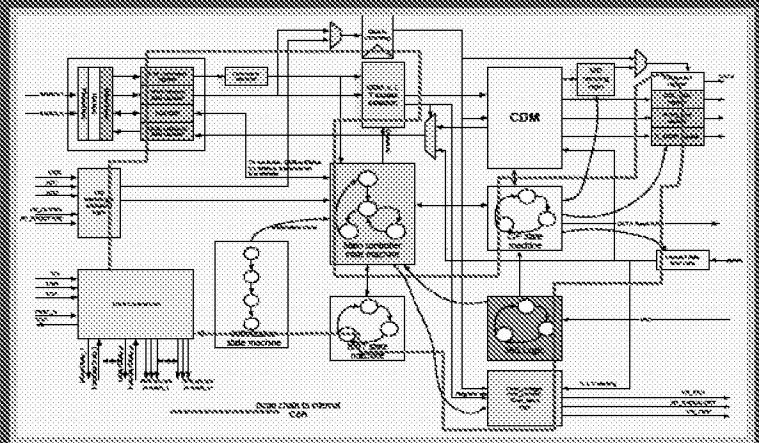


Pulses
16 phases
Variable duty

Triangle Wave
16 phases

Master Control Architecture

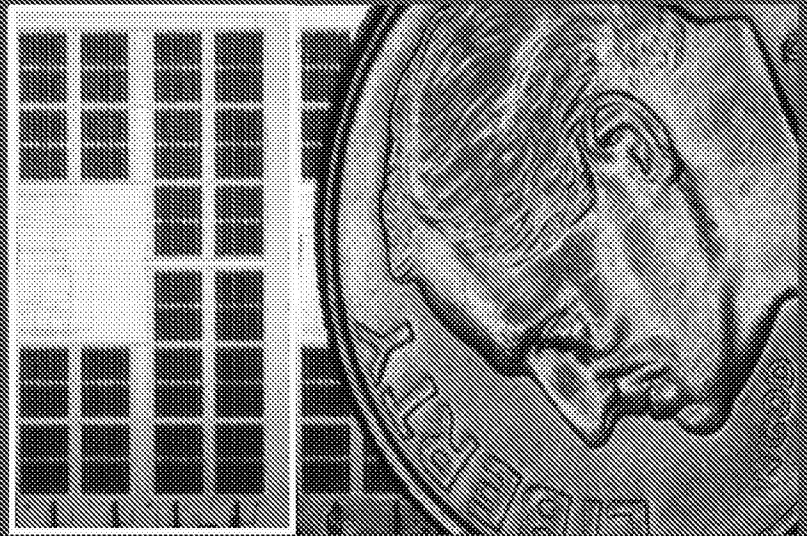
- Master Controller Custom RTL
- VID controller
- JTAG 1194 compliant
- Cell Domain Map (V,T,I)
- AVP adjust
- Test & BIST per cell
- Softstart & Warmstart Algorithms
- Internal Buss interface logic
- IRQ buss
- Platform Interface support
 - Parallel buss
- Cell I-balance



Master Controller Block

Agenda

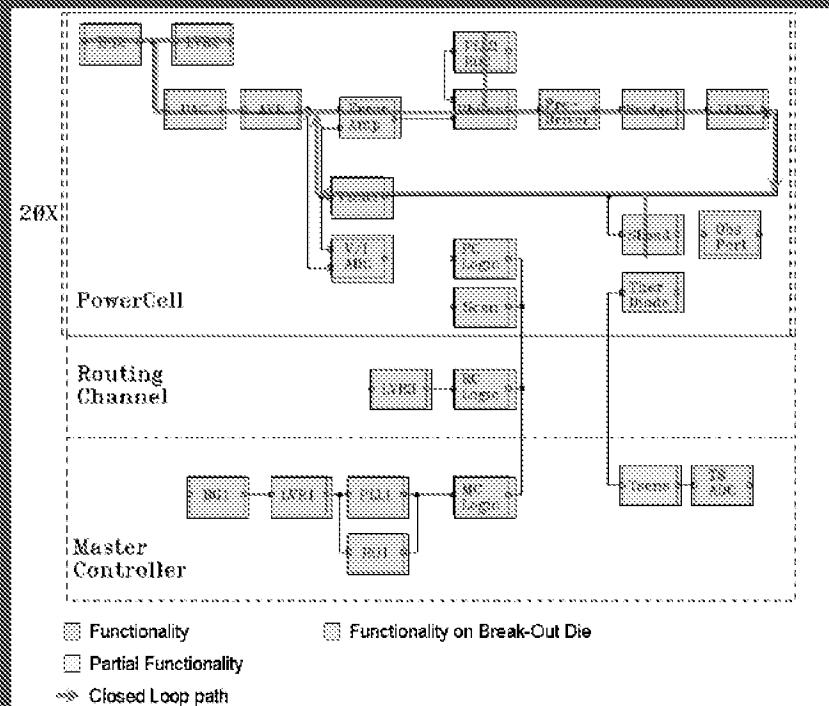
- Architecture
- Circuits
- Results
- Conclusion



Circuit Testing - Schematically

- Validation

- Circuits broken down to analog and digital
- Analog circuits highly observable thru multiple ports
- ADC and DAC conversions for digital readout
- Micro-controller testability thru parallel interface and thru TAP interfaces.



Test Results: Snap-shot of circuits – both Wafer Probe & Package

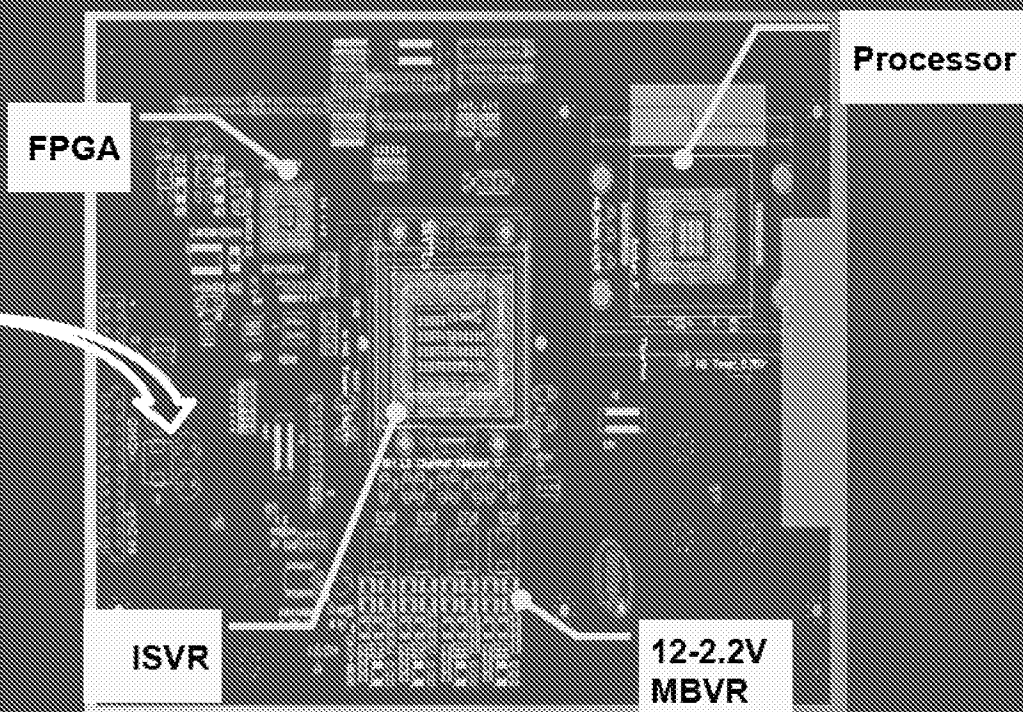
- Bandgap
 - All internal linear regulators (LDO's)
 - Sensors
 - V/I sensors & ADC used for known-good-die screening –all were functional (not fully debugged though)
 - Temperature sensor and its ADC functional on break-out die
 - Interface logic
 - Enabled full programming through either parallel bus or scan
 - DFT features
 - such as manual programming of VCO frequency
 - Observability ports
 - To look at pre-determined internal nodes
- AVP
- Shared all 20 Cells



Platform Testing: Interposer & Microprocessor as load

- For testing with CPU and bench testing

Modified Test Platform
(Used for booting CPU)

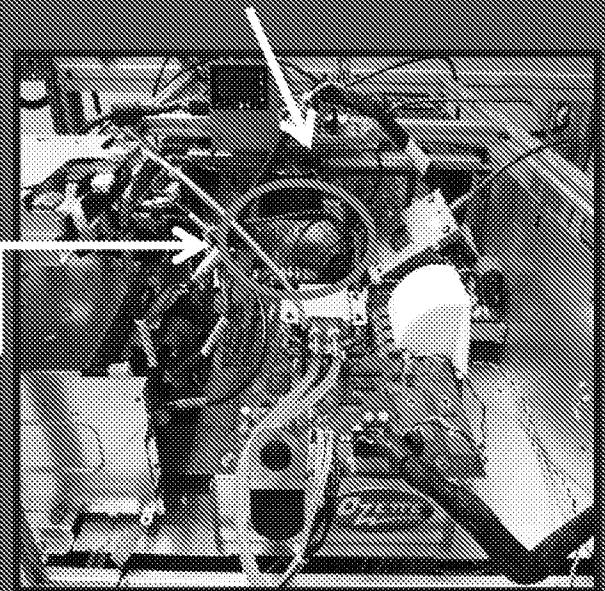
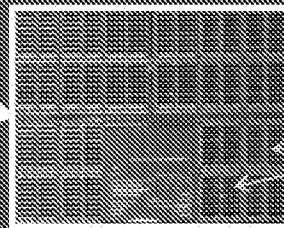
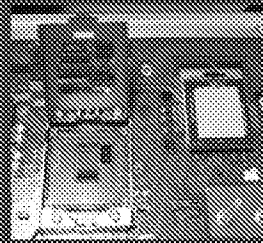
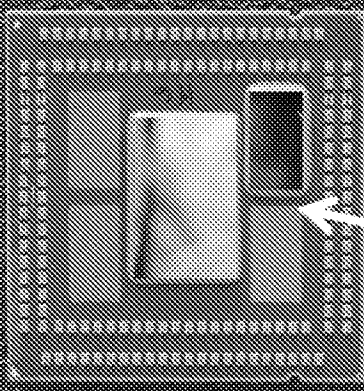


Booted 90W Server CPU – only 3 cells

Intel® Xeon® Processor E7330

- Used only 3 Power Cells
- With 40% of Output Filter Cap*
- Continuous operation with virus for 4+hours
 - No Errors

Ozette



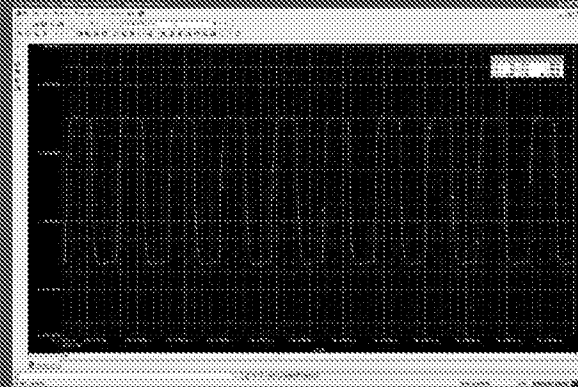
These 3 cells

*Compared with MBVR

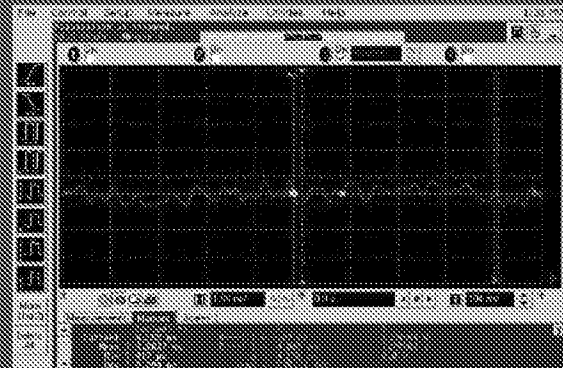


V_{ripple} & V_{TT}

- Voltage Ripple V_{ripple}
- Measurements in lab on ISVR indicate ripple is almost non-existent
- Simulations yielded worst case $\pm 2mV$
- V_{TT} Thermal drift
- Due to thermal time-constant of measurement and error due to linearity circuitry in controller, etc.
- Drift range on package is 60-110C
- Most error is calibrated out and leftover is linearized over temperature range to less than $1mV$



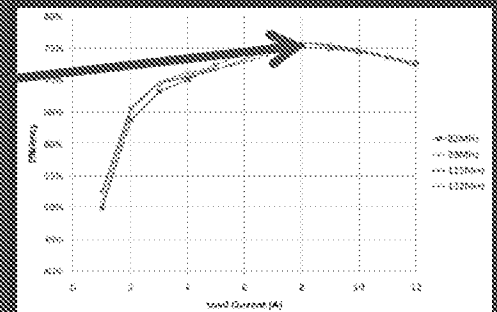
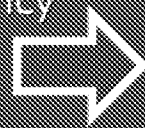
$2mV > V_{ripple}$ sim on Cell



V_{ripple} measurement –
Only noise pickup

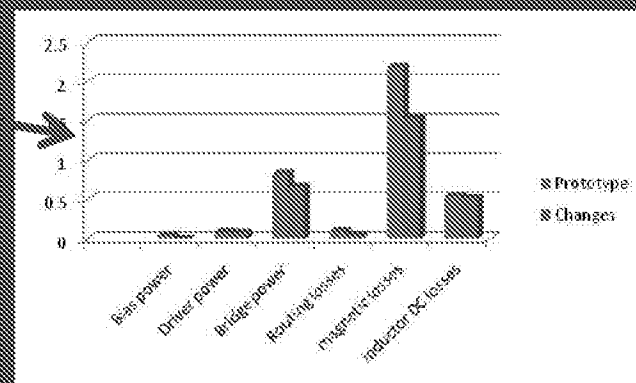
Efficiency – one cell - WIP

- Basic Test (no changes) ~76% peak
 - No bridge shed enable (flat efficiency algorithms not enabled)
 - Bias circuits all on.



'Raw' Efficiency Measurements

- Efficiency* ~82% speculated with basic changes for 'product' level intro
 - Inductor topology coupling change
 - Non-lab level magnetics processed
 - Bias pwr re-distributed
 - Driver/Bridge circuits re-biased
 - Non-test bridge/output routing



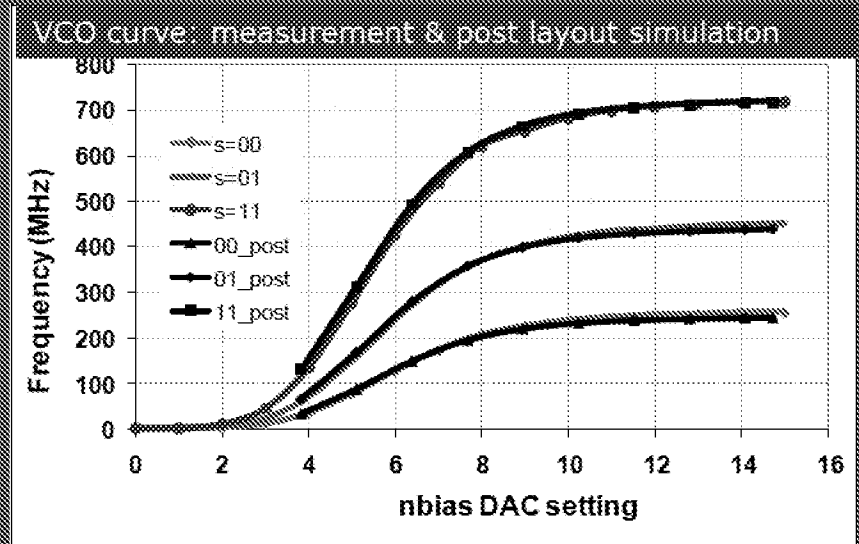
Power Breakdown by element

*Does not include additional advancements that Cannot be reported at this time.



Test Result: Powercell PLL/VCO

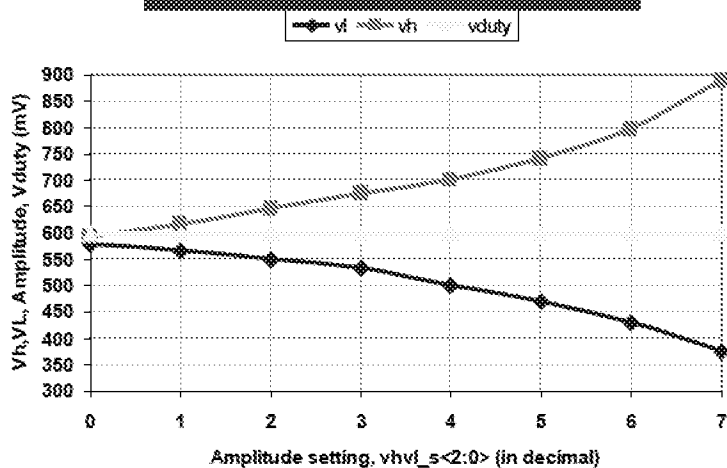
- PLL locking from 20MHz to 200MHz
- Frequencies correlate well with post layout simulation
- Targeted switching frequency, 50-100MHz, is at the linear portion of K_{vco} curve.



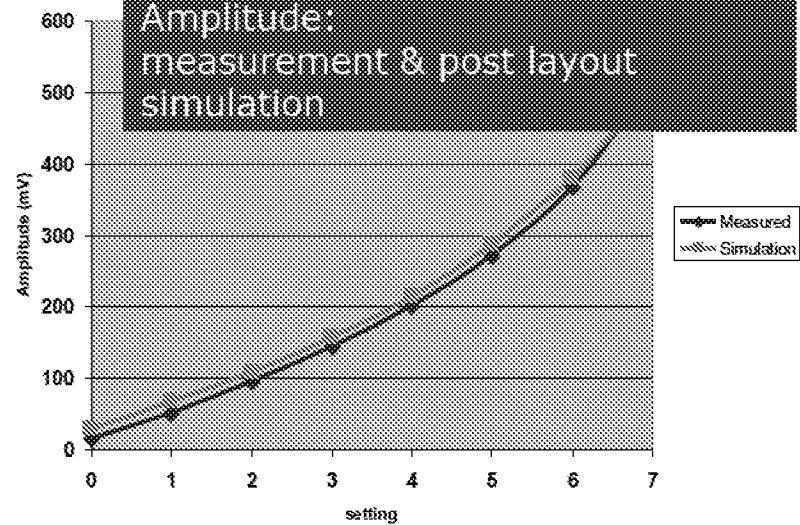
Test Result: Triangle Wave Amplitude

- Triangle Wave amplitude is set by VH/VL control circuit
- A wide range of triangle wave amplitude can be obtained
- Measurement and simulation matched well

VH/VL vs control setting

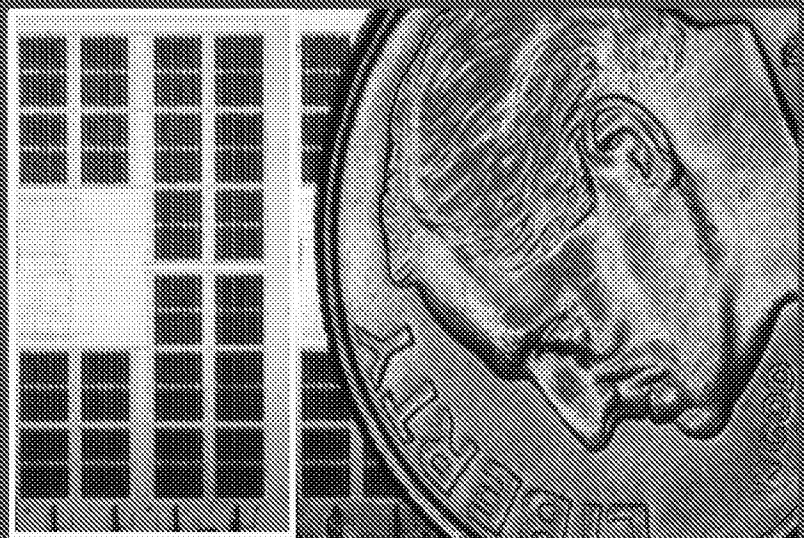


Amplitude:
measurement & post layout
simulation



Agenda

- Architecture
- Circuits
- Results
- Conclusion

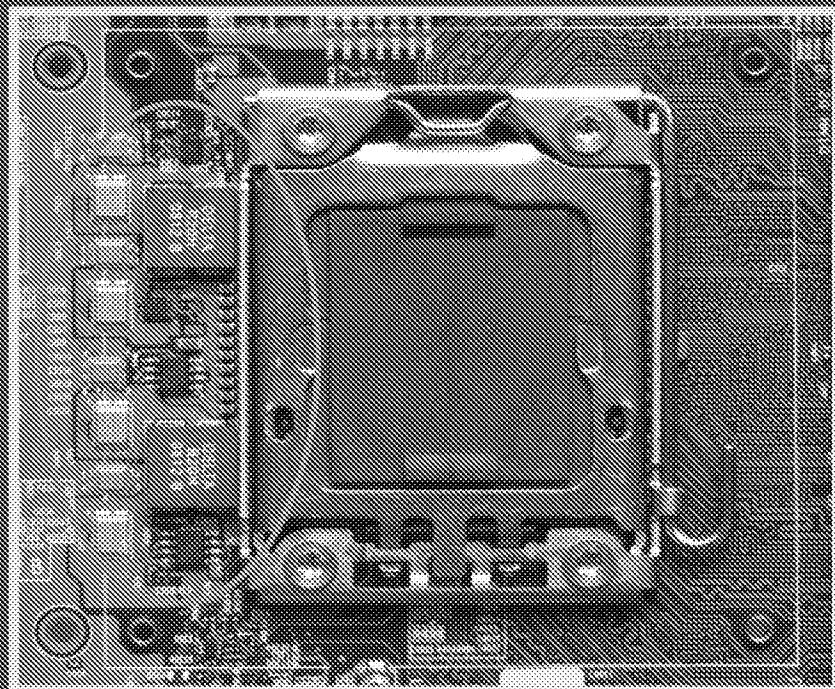


Comparison with other solutions...

- ISVR vs. Platform VR
 - ISVR is $\sim 400A$ design; the other is $\sim 120A$ – input voltages different...but you get the idea.
 - $\sim 110\text{ mm}^2$ vs. 2700 mm^2
- ISVR technology is $\sim 50\times$ smaller


ISVR
 $\sim 12 \times 9$
mm

$\sim 18 \times 50\text{ mm}$
*1 ISVR +
small 12-
2.4V VR
replaces 3.2
of these!*



VR for Intel® Xeon®
Processor E7330



Conclusions

- 400A capable – tested to 220A for less than ½ of chip
 - Board thermally limited.
- Booted and ran server processor (90W design) with 2 cells – ran with 3 cells under Linpack™ for 4+ hours.
- Ripple below noise threshold
- Efficiency in low 80's with minor changes
 - Additional changes possible will boost up.
- Density is ~8A/mm² thermally constrained
- Questions?



June 7, 2017

Attorney Docket No.: 00614-0180001

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Presented for filing is a new continuation in part patent application of:

Inventor(s): PATRIZIO VINCIARELLI and ANDREW T. D'AMICO

Title: METHOD AND APPARATUS FOR DELIVERING POWER TO
SEMICONDUCTORS

This application claims the benefit of the filing date and priority of the following
application(s):

<u>Application No.</u>	<u>Country or Region</u>	<u>Filing Date (MM/DD/YYYY)</u>
15/091,346	United States	04/05/2016

Enclosed are the following papers, including those required to receive a filing date under
37 CFR 1.53(b):

	<u>Pages</u>
Specification	46
Claims	40
Abstract	1
Drawing(s)	30

Enclosures:

- Application Data Sheet, 8 pages.
- Preliminary amendment, 16 pages.
- Information disclosure statement, 1 page.
PTO-1449, 2 page.
- References, 17 item(s).
- Non-Publication Request, 1 page.

Street Address
Fish & Richardson P.C.
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Minneapolis, MN 55440-1022
617 542 5070 main
877 769 7945 fax



Commissioner for Patents
June 7, 2017

Basic Filing Fee			\$280
Search Fee			\$600
Examination Fee			\$720
Total Claims 64	over 20	44 x \$80	\$3520
Independent Claims 6	over 3	3 x \$420	\$1260
Fee for Multiple Dependent claims			\$0
Application size fee for each 50 pages over 100			
Total Sheets: 115x .75 - 100/50 = 0x			\$0
Total Filing fee			\$6380

The total filing fee is being paid concurrently on the Electronic Filing System (EFS) by deposit account authorization. Apply any other required fees or any credits to Deposit Account No. 06-1050, referencing the attorney docket number shown above.

I hereby authorize the USPTO to communicate in response to e-mail communications from the undersigned and practitioners under Customer No. 26214 in accordance with 37 CFR 1.33 and 37 CFR 1.34 concerning any subject matter of this application via e-mail. I understand that a copy of these communications will be made of record in the application file. (MPEP 502.03).

If this application is considered to be incomplete, or if a telephone conference would otherwise be helpful, call the undersigned at (617) 542-5070.

Send all correspondence to:

26214
Customer Number

Respectfully submitted,

/Rex I. Huang/

Rex I. Huang

Registration No. 57,661
Enclosures



Commissioner for Patents
June 7, 2017

RIH/rml
23658266.doc

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Serial No. :		Examiner :	Unknown
Filed :			
Title :	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the references listed on the enclosed PTO-1449 Form. Foreign patent documents and non-patent literature are enclosed; cited U.S. patents and patent application publications will be provided on request. A copy of a communication from a foreign patent office in a counterpart application is also enclosed.

This statement is being filed with the application.

Apply any necessary charges or credits to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: June 7, 2017_____

/Rex I. Huang/
Rex I. Huang
Reg. No. 57,661

Customer Number
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No.
	Applicant VLT, Inc.	
	Filing Date	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	14/731,287		Vinciarelli et al.			06/04/2015
	2.	14/822,561		Vinciarelli			08/10/2015
	3.	14/874,054		Vinciarelli			10/02/2015
	4.	6,930,893	08/16/2005	Vinciarelli			
	5.	6,975,098	12/13/2005	Vinciarelli			
	6.	6,984,965	01/10/2006	Vinciarelli			
	7.	2013/0069608	03-2013	Gakhar et al.			
	8.	9,112,422	08/18/2015	Vinciarelli			
	9.	9,166,481	10/15/2015	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	10.	JPH02280666	11/16/1990	JP			English Abstract	
	11.	EP2863531	4/22/2015	EP				

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	12.	DiBene II et al., "A 400 Amp Fully Integrated Silicon Voltage Regulator with in-die Magnetically Coupled Embedded Inductors," retrieved from http://www.psma.com/sites/default/files/uploads/tech-forums-nanotechnology/resources/400a-fully-integrated-silicon-voltage-regulator.pdf , retrieved on August 3, 2016, 25 pages.
	13.	European Office Communication in European Application No. 16173089.0, dated April 10, 2017, 2 pages
	14.	Extended Search Report in European Application No. 16173089.0, dated February 14, 2017, 8 pages
	15.	Miller, "12V PowerStage in Embedded Die System-in-Package," APEC 2015, 33 pages
	16.	Okudaira and Matsuse, "A New Quasi-Resonant Inverter With Two-way Short-circuit Switch Across A Resonant Capacitor," Power Conversion Conference 2002, 1496-1501
	17.	Watson, "New Techniques in the Design of Distributed Power Systems," Dissertation, Virginia Polytechnic Institute and State University, 1998, 32 pages

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0180001
		Application Number	
Title of Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Patrizio		Vinciarelli		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Boston	State/Province	MA	Country of Residence	US

Mailing Address of Inventor:

Address 1	294 Beacon Street				
Address 2					
City	Boston	State/Province	MA		
Postal Code	02116	Country	US		

Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Andrew	T.	D'Amico		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Beverly Hills	State/Province	CA	Country of Residence	US

Mailing Address of Inventor:

Address 1	3046 Franklin Canyon Drive				
Address 2					
City	Beverly Hills	State/Province	CA		
Postal Code	90210	Country	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0180001
		Application Number	
Title of Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

☐ An Address is being provided for the correspondence information of this application.

Customer Number	26214		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		
Attorney Docket Number	00614-0180001	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	30	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☒ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	26214		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0180001
		Application Number	
Title of Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Continuation in part of	15091346	2016-04-05
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0180001
		Application Number	
Title of Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0180001
		Application Number	
Title of Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	VLT, Inc.	
Mailing Address Information For Applicant:		
Address 1	25 Frontage Road	
Address 2		
City	Andover	State/Province
Country	US	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0180001
		Application Number	
Title of Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

Assignee	1			
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Signature:[Remove](#)

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Rex I. Huang/		Date (YYYY-MM-DD)	2017-06-07
First Name	Rex	Last Name	Huang	Registration Number
				57661
Additional Signature may be generated within this form by selecting the Add button. Add				

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0180001
		Application Number	
Title of Invention	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

NONPUBLICATION REQUEST UNDER 35 U.S.C. 122(b)(2)(B)(i)	First Named Inventor		Patrizio Vinciarelli
	Title	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS	
	Attorney Docket Number		00614-0180001

I hereby certify that the invention disclosed in the attached application **has not and will not be** the subject of an application filed in another country, or under a multilateral agreement, that requires publication at eighteen months after filing.

I hereby request that the attached application not be published under 35 U.S.C. 122(b).

/Rex I. Huang/	June 7, 2017
_____ Signature	_____ Date
Rex I. Huang	57,661
_____ Typed or printed name	_____ Registration Number, if applicable
617-542-5070	
_____ Telephone Number	

This request must be signed in compliance with 37 CFR 1.33(b) and submitted with the application **upon filing**.

Applicant may rescind this nonpublication request at any time. If applicant rescinds a request that an application not be published under 35 U.S.C. 122(b), the application will be scheduled for publication at eighteen months from the earliest claimed filing date for which a benefit is claimed.

If applicant subsequently files an application directed to the invention disclosed in the attached application in another country, or under a multilateral international agreement, that requires publication of applications eighteen months after filing, the applicant **must** notify the United States Patent and Trademark Office of such filing within forty-five (45) days after the date of the filing of such foreign or international application. **Failure to do so will result in abandonment of this application (35 U.S.C. 122(b)(2)(B)(iii)).**

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 06/14/2017

SDUONG	ADJ #00000001	Mailroom Dt: 06/07/2017		
	Seq No: 1620	Sales Acctg Dt: 06/08/2017	061050	15616288
	04 FC : 1202	3520.00 CR		

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 06/14/2017

SDUONG	SALE	#00000004	Mailroom Dt:	06/07/2017	061050	15616288
		01	FC : 1051	140.00	DA	

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 06/14/2017

SDUONG	SALE	#00000017	Mailroom Dt:	06/07/2017	061050	15616288
		01	FC : 1202	3,440.00	DA	



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001

CONFIRMATION NO. 7438

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

INFORMAL NOTICE



Date Mailed: 06/20/2017

INFORMATIONAL NOTICE TO APPLICANT

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

The item(s) indicated below are also required and should be submitted with any reply to this notice to avoid further processing delays.

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):
Patrizio Vinciarelli
Andrew T. D'Amico

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/wtsige/

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
15/616,288

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	63 minus 20 =	* 43
INDEPENDENT CLAIMS (37 CFR 1.16(h))	6 minus 3 =	* 3
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	280
N/A	600
N/A	720
x 80 =	3440
x 420 =	1260
	0.00
	0.00
TOTAL	6300

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*		Minus	**	=
Independent (37 CFR 1.16(h))	*		Minus	***	=
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*		Minus	**	=
Independent (37 CFR 1.16(h))	*		Minus	***	=
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
15/616,288	06/07/2017	2827	6440	00614-0180001	63	6

CONFIRMATION NO. 7438

FILING RECEIPT

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022



Date Mailed: 06/20/2017

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Patrizio Vinciarelli, Boston, MA;
Andrew T. D'Amico, Beverly Hills, CA;

Applicant(s)

VLT, Inc., Andover, MA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CIP of 15/091,346 04/05/2016

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 06/14/2017

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 15/616,288**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

Preliminary Class

365

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Jun 20, 2017 03:28:40 AM

Dear PAIR Customer:

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 26214 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

Disclaimer:

The list of documents shown below is provided as a courtesy and is not part of the official file wrapper. The content of the images shown in PAIR is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
15616288	M327	06/20/2017	00614-0180001
	APP.FILE.REC	06/20/2017	00614-0180001

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

SCORE Placeholder Sheet for IFW Content

Application Number: 15616288

Document Date: 06/07/2017

The presence of this form in the IFW record indicates that the following document type was received in electronic format on the date identified above. This content is stored in the SCORE database.

- Drawings – Other than Black and White Line Drawings

Since this was an electronic submission, there is no physical artifact folder, no artifact folder is recorded in PALM, and no paper documents or physical media exist. The TIFF images in the IFW record were created from the original documents that are stored in SCORE.

To access the documents in the SCORE database, refer to instructions below.

At the time of document entry (noted above):

- Examiners may access SCORE content via the eDAN interface.
- Other USPTO employees can bookmark the current SCORE URL (<http://Score.uspto.gov/ScoreAccessWeb/>).
- External customers may access SCORE content via the Public and Private PAIR interfaces.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date
15/616,288	6/7/2017

(Note: The boxes above may be left blank if information is provided on the Transmittal.)

- ☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter, which may be an application filed after the date of this power of attorney, or identified above:

26214

OR

- ☐ I hereby appoint Practitioner(s) named in the attached list as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter, which may be an application filed after the date of this power of attorney, or identified above.

Please recognize or change the correspondence address for the application identified in the attached transmittal letter to:

- ☐ The address associated with the above-mentioned Customer Number

OR

- ☒ The address associated with Customer Number: **26214**

OR

Firm or Individual Name				
Address				
City		State		Zip
Country				
Telephone		Email		

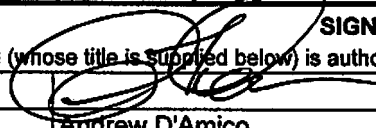
I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

VLT, Inc.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature		Date (Optional)	2017-06-27
Name	Andrew D'Amico		
Title	V.P., Intellectual Property		

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

- ☐ *Total of _____ forms are submitted.

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)**Title of
Invention****METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS**

As the below named inventor, I hereby declare that:

This declaration [] The attached application, or
is directed to:[X] United States application or PCT international application
number15/616,288filed on June 7, 2017.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001
by fine or imprisonment of not more than five (5) years, or both.**WARNING:**

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTORInventor: Patrizio Vinciarelli

Date (Optional):

June 27, 2017Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)**Title of
Invention****METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS**

As the below named inventor, I hereby declare that:

This declaration ☐ The attached application, or
is directed to:☒ United States application or PCT international application
number15/616,288filed on June 7, 2017.

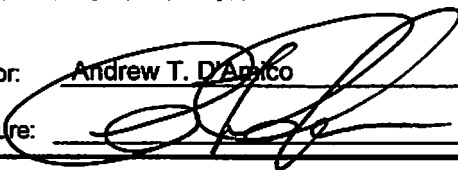
The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTORInventor: Andrew T. D'AmicoDate (Optional): 2017-06-27Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Patent Application Fee Transmittal

Application Number:	15616288			
Filing Date:	07-Jun-2017			
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0180001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
LATE FILING FEE FOR OATH OR DECLARATION	1051	1	140	140
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				140

Electronic Acknowledgement Receipt

EFS ID:	29636068
Application Number:	15616288
International Application Number:	
Confirmation Number:	7438
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0180001
Receipt Date:	28-JUN-2017
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Application Type:	Utility under 35 USC 111(a)

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0180001_Trans_POA.pdf	127745	yes	2
			fb2b1fdfe656cb05f8cf19ad4640304289e893c		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal Letter		1	1	
	Power of Attorney		2	2	
Warnings:					
Information:					
2	Oath or Declaration filed	00614_0180001_PV_Dec_Signed.pdf	290035	no	1
			e6408f17f3db7e5d6834c0bda12deec0b96f41f1		
Warnings:					
Information:					
3	Oath or Declaration filed	00614_0180001_ATD_Dec_Signed.pdf	70957	no	1
			d68b465b6e7f1ce3c07e90bd10180e6e18bd5fe97		
Warnings:					
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4	Fee Worksheet (SB06)	fee-info.pdf	30271	no	2
			a549b6c319ba6d829ca61e906924e0be6be60fa5		
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

Application Number	15/616,288
Filing Date	June 7, 2017
First Named Inventor	Patrizio Vinciarelli
Title	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS
Art Unit	
Examiner Name	
Attorney Docket Number	00614-0180001

SIGNATURE of Applicant or Patent Practitioner

Signature	/Rex I. Huang/	Date (Optional)	
Name	Rex I. Huang	Registration Number	57,661
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ *Total of _____ forms are submitted.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001

CONFIRMATION NO. 7438

POA ACCEPTANCE LETTER

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022



Date Mailed: 06/30/2017

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 06/28/2017.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/mnguyen/

To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
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Jun 30, 2017 10:08:52 AM

Dear PAIR Customer:

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 26214 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

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Application	Document	Mailroom Date	Attorney Docket No.
15616288	N570	06/30/2017	00614-0180001

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Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Serial No. :	15/616,288	Examiner :	Unknown
Filed :	June 7, 2017	Conf. No. :	7438
Title :	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the references listed on the enclosed PTO-1449 Form. Non-patent literature is enclosed; cited U.S. patents and patent application publications will be provided on request. Copies of communications from the patent office in related applications are also enclosed.

This statement is being filed before the receipt of a first action on the merits.

Apply any necessary charges or credits to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: August 1, 2017_____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0180001	Application No. 15/616,288
Information Disclosure Statement by Applicant (Use several sheets if necessary)	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit
(37 CFR §1.98(b))		

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	6,031,743	02/29/2000	Carpenter et al.			
	2.	6,035,261	03/07/2000	Carpenter et al.			
	3.	6,275,958	08/14/2001	Carpenter et al.			
	4.	6,300,749	10/01/2001	Castelli et al.			
	5.	6,421,262	07/16/2002	Saxelby et al.			
	6.	6,466,458	10/15/2002	Zhang et al.			
	7.	6,603,292	08/01/2003	Schouten et al.			
	8.	6,700,361	03/01/2004	Gregorius, Peter			
	9.	6,788,033	09/07/2004	Vinciarelli			
	10.	6,911,848	06/28/2005	Vinciarelli			
	11.	6,934,166	08/23/2005	Vinciarelli			
	12.	6,969,909	11/29/2005	Briere			
	13.	6,985,341	01/10/2006	Vinciarelli, et al.			
	14.	7,145,786	12/05/2006	Vinciarelli			
	15.	7,521,909	04/01/2009	Dow et al.			
	16.	7,561,446	07/14/2009	Vinciarelli			
	17.	7,768,807	08/01/2010	Chen, et al.			
	18.	8,232,783	07/01/2012	Yanagawa, Kenji			
	19.	RE40,072	02/19/2008	Prager, et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	20.	Amendment after Allowance Pursuant to 37 C.F.R. §1.312 in U.S. Application No. 13/044,063, dated March 3, 2015, 17 pages

Examiner Signature	Date Considered
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0180001	Application No. 15/616,288
Information Disclosure Statement by Applicant (Use several sheets if necessary)		Applicant VLT, Inc.	
		Filing Date June 7, 2017	Group Art Unit
(37 CFR §1.98(b))			

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	21.	AN2738, Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009
	22.	Burton et al., "FIVR - Fully Integrated Voltage Regulators on 4th Generation Intel ® CoreTM SoCs," Applied Power Electronics Conference and Exposition (APEC), 2014 Twenty-Ninth Annual IEEE, March 16, 2014, 432-439
	23.	ISL6210, Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008
	24.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated February 20, 2015, 15 pages
	25.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated June 5, 2015, 11 pages
	26.	Notice of Allowance in U.S. Application No. 14/682,187, dated January 17, 2017, 22 pages
	27.	Office Action in U.S. Application No. 13/044,063, Examiner Nguyen, dated May 23, 2014, 24 pages
	28.	Reply to Action of May 23, 2014 in U.S. Application No. 13/044,063, dated October 24, 2014, 39 pages
	29.	Supplemental Notice of Allowability in U.S. Application No. 13/044,063, Examiner Nguyen, dated April 20, 2015, 5 pages

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2838
Serial No. :	13/044,063	Examiner :	Jye-June Lee
Filed :	March 9, 2011	Confirmation No. :	1081
		Notice of Allowance Date:	2/20/2015
Title :	FAULT TOLERANT POWER CONVERTER		

MAIL STOP ISSUE FEE

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AFTER ALLOWANCE
PURSUANT TO 37 C.F.R. §1.312

Please amend the application as indicated on the following pages. This amendment is being filed prior to payment of the issue fee.

Amendments to the Claims:

This listing of claims replaces all prior versions and listings of claims in the application:

Listing of Claims:

1. (Currently Amended) A method of converting power via a transformer for supplying a low voltage output comprising:

providing a secondary winding having a first terminal and a second terminal;

providing a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction;

connecting the first series circuit across an output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the ~~second~~ first switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

2. (Previously Presented) The method of claim 1 further comprising:

sensing across each of the first and second switches, a respective voltage polarity; and

turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

3. (Previously Presented) The method of claim 1 further comprising:
providing a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the second terminal of the secondary winding to the second junction; and

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

4. (Previously Presented) The method of claim 3 further comprising:

preventing the third switch from turning ON if the fourth switch fails to open;

preventing the fourth switch from turning ON if the third switch fails to open.

5. (Previously Presented) The method of claim 4 further comprising:

sensing across each of the third and fourth switches, a respective voltage polarity; and

turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

6. (Previously Presented) The method of claim 5 further comprising:

providing a first control circuit to operate the first switch;

providing power to operate the first control circuit and first switch from across the third switch;

providing a third control circuit to operate the third switch; and